

Multi-Copy Security in Unclonable Cryptography

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Abstract

Unclonable cryptography leverages the quantum no-cloning principle to copy-protect cryptographic functionalities. While most existing works address the basic single-copy security, the stronger notion of multi-copy security remains largely unexplored.

We introduce a generic compiler that upgrades collusion-resistant unclonable primitives to achieve multi-copy security, assuming only one-way functions. Using this framework, we obtain the first multi-copy secure constructions of public-key quantum money (termed quantum coins), single-decryptor encryption, unclonable encryption, and more. We also introduce an extended notion of quantum coins, called upgradable quantum coins, which allow weak (almost-public) verification under weaker assumptions and can be upgraded to full public verification under stronger assumptions by the bank simply publishing additional classical information.

Along the way, we give a generic compiler that upgrades single-copy secure single-decryptor encryption to a collusion-resistant one, assuming the existence of functional encryption, and construct the first multi-challenge secure unclonable encryption scheme, which we believe are of independent interest.

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1 Introduction

The no-cloning theorem is a fundamental and distinctive feature of quantum information theory. It asserts that it is impossible to create an identical copy of an arbitrary unknown quantum state. This stands in stark contrast to classical information, which can be copied freely. The no-cloning theorem has had profound implications in the field of cryptography. One of the earliest applications is quantum money [Wie83], whose authenticity can be verified but which cannot be cloned. Building on this idea, a wide range of unclonable cryptographic primitives have been explored, including copy-protection [Aar09], unclonable encryption [BL20], single-decryptor encryption [GZ20, CLLZ21], secure leasing [AL21], certified deletion [BI20], and more.

Multi-copy security. In studies of unclonable cryptography, we typically focus on the $1 \rightarrow 2$ unclonability setting, where an adversary is given a single copy of a pure quantum state but is prohibited from producing two copies. In more realistic scenarios, however, one may also consider the more general $q \rightarrow q+1$ setting, in which the adversary is provided with q copies of a pure state but cannot produce $q+1$ copies. We refer to this property as *multi-copy security*. Throughout the paper, we focus on unbounded multi-copy security, where q is an arbitrary, a priori unbounded polynomial unless otherwise stated.

The notion of multi-copy security has been investigated in several works. However, the known feasibility results for achieving multi-copy security remain quite limited. To the best of our knowledge, only the following results address multi-copy security in the context of unclonable cryptography:

- Mosca and Stebila [MS10] introduced the notion of *quantum coins*, which is a multi-copy secure version of public-key quantum money. In a quantum coin scheme, a “coin” is a pure state, and security requires that an adversary given q exact copies of the coin cannot produce $q+1$ valid coins. They constructed a quantum coin scheme in the quantum oracle model and left open the problem of constructing a quantum coin scheme in the plain model.
- Ji, Liu, and Song [JLS18] constructed a multi-copy secure private-key quantum money scheme based on pseudorandom quantum states.
- Ananth, Mutreja, and Poremba [AMP25] constructed a (one-time secret key) encryption scheme that satisfies an oracular notion of multi-copy certified deletion security based on one-way functions. Note that their oracular notion of security is weaker than the standard certified deletion security (see Section 1.2 for the details). They also constructed a multi-copy secure leasing scheme for general programs in a classical oracle model.
- Poremba, Ragavan, and Vaikuntanathan [PRV24] constructed a (one-time secret key) unclonable encryption (UE) scheme that satisfies an oracular notion of bounded multi-copy security. Again, their oracular security notion is weaker than the standard security of UE (see Section 1.2 for the details).

Despite these advances, many fundamental challenges in understanding and constructing multi-copy secure cryptographic schemes remain largely unexplored. For example, constructing a quantum coin scheme in the plain model and improving the results of [AMP25, PRV24] to achieve standard (non-oracular) security and to achieve unbounded multi-copy security (in the case of UE) remain open problems. Furthermore, the multi-copy security of other unclonable primitives has not yet been investigated.

Collusion-resistance. A notion closely related to multi-copy security is collusion resistance. In this setting, the adversary is given multiple quantum states that are independently generated by repeatedly executing the same generation algorithm. This stands in stark contrast to multi-copy security, where the adversary receives multiple exact copies of the same pure state.

The collusion-resistance of unclonable cryptographic primitives is better understood than their multi-copy security counterparts, and several constructions have been recently proposed. For example, Aaronson and Christiano [AC13] (based on earlier ideas of [LAF⁺09, FGH⁺12]) showed that a single-copy secure public-key quantum money scheme (often referred to as a mini-scheme) can be generically upgraded to a collusion-resistant secure scheme by combining it with a digital signature scheme. In addition, [CG24] constructed (using ideas from the bounded-collusion scheme of [LLQZ22]) a collusion-resistant single-decryptor encryption (SDE) schemes and [KNP25] constructed a collusion-resistant secure key leasing (SKL) scheme for public-key encryption (PKE) and, more generally, for attribute-based encryption (ABE) based on standard assumptions in the plain model.

While collusion-resistance and multi-copy security are conceptually similar, the multi-copy secure counterparts of the above schemes remain unknown. This naturally raises the following question:

Is there a way to bridge the gap between collusion-resistance and multi-copy security?

Why multi-copy security? A natural question is why one should pursue multi-copy security instead of collusion resistance. In fact, the states considered in a collusion-resistant setting can be interpreted as multiple copies of mixed states that account for the randomness of the generation algorithm. In what follows, we outline several reasons for focusing on multi-copy security.

First, consider the operational advantage. Equality of pure states can be efficiently tested using the SWAP test (up to some bounded error), and this capability can be useful in applications. For instance, in the context of secure key leasing, where decryption keys are encoded as quantum states, multi-copy security allows one to compare multiple leased keys for equality. [AMP25] argued that such a capability could lead to the notion of nested key leasing, where a leased key can itself be further leased to another party.

Second, consider the aspect of anonymity as discussed in [MS10]. When identical copies of the same pure state are distributed, there is no inherent way to distinguish among them, naturally giving rise to a form of anonymity. (See [Section 1.2](#) for more discussions).

Finally, there is also a conceptual motivation. While multiple copies of a pure state correspond to identical physical objects, states sampled from the same distribution need not be identical, they may be distinct states generated according to the same preparation procedure. In a classical analogy, if two strings x and y are drawn from the same distribution but take different values, we would not regard them as “copies.” Hence, multi-copy security more faithfully captures the intuition of having exact duplicates in the quantum setting.

1.1 Our Results

We present a generic compiler that upgrades collusion-resistance into multi-copy security for a broad class of unclonable cryptographic primitives, assuming the existence of one-way functions (OWFs). Namely, we prove the following theorem.

Theorem 1 (Informal). *Suppose that there is a QPT algorithm GenState that takes a classical input z and a classical randomness $\text{rand} \in \{0, 1\}^r$, and outputs a pure quantum state $|\phi_{z, \text{rand}}\rangle$ fully*

determined by z and rand . For a pseudorandom state (PRS) key k for λ -qubit output states and a pseudorandom function (PRF) key K for a function F from $\{0, 1\}^\lambda$ to $\{0, 1\}^r$, define $|\psi_{z,k,K}\rangle = \sum_x \alpha_{k,x} |x\rangle \otimes |\phi_{z,F(K,x)}\rangle$ where $\sum_x \alpha_{k,x} |x\rangle$ is the state produced by the PRS scheme with the key k . Then, for any z and any polynomial t , given $|\phi_{z,\text{rand}_1}\rangle \otimes \dots \otimes |\phi_{z,\text{rand}_t}\rangle$ for independently random $\text{rand}_1, \dots, \text{rand}_t$, one can efficiently generate a state that is computationally indistinguishable from $|\psi_{z,k,K}\rangle^{\otimes t}$ for uniformly random k, K .

The formal version of the above theorem is given in [Theorem 7](#). The theorem provides a way to switch between t independently generated states to t copies of a state produced with the same randomness. This establishes a compiler that compiles collusion-resistance into multi-copy security.

To apply our compiler, we have to assume that the algorithm generating a copy-protected quantum state satisfies the assumption for **GenState** in the above theorem; namely, its output must be a pure state fully determined by the input and the classical randomness. We say that such a quantum algorithm has *classically determined outputs*. Fortunately, many existing constructions of unclonable primitives satisfy this property, and thus our compiler is applicable to many existing constructions.

1.1.1 Quantum Coins

In [Section 6.1](#), we first apply our compiler to quantum money schemes. It is known that there exists a collusion-resistant public-key quantum money scheme based on subspace-hiding obfuscation¹ and OWFs [\[AC13, Zha19b\]](#). We observe that the banknote generation algorithm in these schemes has classically determined outputs. Therefore, we can apply our compiler to obtain quantum coins based on subspace-hiding obfuscation and OWFs. This yields the first construction of quantum coins in the plain model, resolving an open problem posed by [\[MS10\]](#). In fact, our construction is generic, being based on any public-key mini-scheme [\[AC13\]](#) whose banknote generation algorithm has classically determined outputs, together with any digital signature scheme.

Separately, in [Section 6.2](#), for the first time, we also prove the security of the folklore quantum coin scheme that simply uses an equal superposition of all possible mini-scheme banknotes. While this needs additional assumptions compared to our pseudorandom state based compiler, we believe this still might be of independent theoretical interest. Additionally, we develop two tools ([Section 4](#) and [Section 5](#)) to prove the security of this scheme, and these tools might have independent applications for quantum cryptography with pure states.

Upgradable quantum coins. We extend our construction of quantum coins to what we call upgradable quantum coins. Upgradable quantum coins are a new notion in which the security of the quantum money scheme holds under a weaker assumption (e.g., public-key encryption), but only in the almost-public setting [\[BS21\]](#), where verification of a banknote is done by comparing it with other banknotes that are already known to be correct. Once an implementation of a stronger cryptographic scheme is available, it can be upgraded to a fully public-key quantum money scheme by the bank releasing a single classical value. We construct upgradable quantum coins whose security before the release (as almost-public coins) holds under the assumption of subexponentially secure public-key encryption, and whose security after the release (as public-key quantum money) holds under the assumptions of subexponentially secure public-key encryption and subspace-hiding obfuscation.

¹This is implied by indistinguishability obfuscation and one-way functions [\[Zha19b\]](#).

1.1.2 Single-Decryptor Encryption.

A single-decryptor encryption (SDE) [GZ20, CLLZ21] is a public key encryption scheme where decryption keys can be copy-protected. A prior work [CG24] constructed collusion-resistant SDE schemes based on subexponentially secure indistinguishability obfuscation (iO) and the LWE assumption. Thus, one can directly apply our compiler to obtain a multi-copy secure variant under these assumptions. However, this construction does not appear to be optimal in terms of assumptions. A recent work [KY25] constructed a single-copy secure SDE scheme under only polynomially secure iO and OWFs, and thus we hope to achieve collusion-resistance and multi-copy security under the same assumptions as well.

To this end, we design a generic compiler that upgrades any single-copy secure SDE scheme into a collusion-resistant one using functional encryption (FE) as a building block. By applying this compiler to the single-copy secure scheme of [KY25], we obtain a collusion-resistant SDE under the same assumption. Finally, by applying our main compiler, this scheme can be further transformed into a multi-copy secure SDE.

We believe that our generic compiler from single-copy security to collusion resistance for SDE is of independent interest, since existing works on (even bounded) collusion-resistant SDE [LLQZ22, CG24] rely on specific structural properties of the single-copy secure SDE scheme of [CLLZ21], rather than treating it as a black-box.

1.1.3 Unclonable Encryption

Unclonable encryption (UE) [BL20] is a one-time secret-key encryption scheme in which each ciphertext is copy-protected. In the single-copy setting, UE schemes achieving search-based security can be constructed unconditionally; however, no construction with indistinguishability-based security is known in the plain model. Since this work focuses on constructions in the plain model, we concentrate on search-based security.

Our compiler reduces the construction of a multi-copy secure UE scheme to that of a multi-challenge secure one, where security holds even when the adversary is given multiple ciphertexts of the same random message, each generated under independent encryption randomness.² However, no construction of a multi-challenge secure UE scheme is currently known. The main difficulty in achieving this lies in the fact that the hybrid argument is not applicable in the UE setting, as there are two non-communicating adversaries. As a result, we cannot perform a simple reduction to the single-challenge setting, as is done in standard (reusable) encryption schemes.

We overcome this challenge by connecting multi-challenge secure UE to collusion-resistant SDE. We observe that if we have an SDE scheme that satisfies collusion-resistant identical-challenge security, where all second-stage adversaries receive identical challenge ciphertexts, then a straightforward conversion from SDE to UE, in which we simply switch the roles of ciphertexts and keys, yields a multi-challenge secure UE scheme. Fortunately, we show that an SDE scheme obtained from our single-copy-to-collusion-resistant compiler also satisfies collusion-resistant identical-challenge security. As a result, we obtain a multi-challenge secure UE scheme assuming the existence of polynomially secure iO and OWFs. We emphasize that this is the first construction of a multi-challenge secure UE scheme under any assumption, which we believe is of independent interest.

Finally, by applying our compiler, we also obtain a multi-copy secure UE scheme under the same assumptions. This improves upon the result of [PRV24] that achieved a weaker bounded multi-copy and oracular notion of security, albeit under a stronger assumption.

²We use the term multi-challenge rather than collusion-resistant, as we believe it is more suitable in the context of UE.

1.1.4 Other Applications

Beyond those seen so far, our compiler would be applicable to a wide range of unclonable cryptographic primitives. We briefly review some examples below.

- By applying our compiler to the collusion-resistant PKE and ABE schemes with secure key leasing (SKL) from [KNP25], we obtain their multi-copy secure counterparts based on the LWE assumption. This yields the first construction of multi-copy secure SKL for any type of encryption scheme.
- In the setting of certified deletion, multi-challenge security can be straightforwardly reduced to single-challenge security via a hybrid argument, provided that the underlying scheme supports reusable security, i.e., it provides an encryption oracle to the adversary (a condition automatically satisfied in the public-key setting). Thus, by applying our compiler to the constructions of encryption with publicly verifiable certified deletion from [KNY23, BKM⁺23], we obtain multi-copy secure reusable secret-key (resp. public-key) encryption schemes with publicly verifiable certified deletion based on OWFs (resp. PKE). This would improve upon the result of [AMP25], which achieved only a weaker oracular notion of security. More generally, our compiler applies to any type of encryption considered in [KNY23, BKM⁺23]. Note that our compiler does not preserve *everlasting security*; however, [AMP25] does not consider everlasting security either.
- Untelegraphable encryption (UTE) [CKNY25] is a weaker variant of UE in which the security requirement only demands that a quantum ciphertext cannot be converted into a classical string that could later be used to recover the message given the decryption key. [CKNY25] constructed a multi-challenge secure UTE scheme³ based on any (possibly quantum) reusable secret-key encryption scheme. By applying our compiler to this construction, we obtain its multi-copy secure counterpart under the assumption that OWFs exist. This yields the first construction of a multi-copy secure UTE scheme.

We omit the details of the above constructions, as they are obtained in a straightforward manner by applying our compiler to existing constructions.

1.2 Related Work

Anonymous public key quantum money. While the notion of quantum coins was originally motivated by anonymity, a recent work [CGY24] constructed anonymous public-key quantum money without relying on quantum coins. In fact, our quantum coin scheme provides a weaker anonymity guarantee compared to theirs. In a quantum coin scheme, all (honestly generated) banknotes are identical pure states. Therefore, if the bank honestly issues the banknotes and adversaries do not tamper with the banknotes, perfect anonymity is satisfied. Thus, our scheme satisfies a semi-honest anonymity notion where the adversaries are trying to track banknotes by recording information without tampering with the banknotes. However, a malicious bank or user might attempt to tamper with the money state to embed distinguishing information while still ensuring that the modified state passes verification. Such an attack would allow the adversary to trace the movement of the state, thereby breaking anonymity. A possible countermeasure is to use SWAP test to compare one's banknote to some banknotes that are somehow known to be honest. A similar approach was used by [BS21]. Note that this is not ideal, since it might be unrealistic

³They refer to it as collusion-resistant UTE, but we use the term multi-challenge secure UTE for consistency with our notion of multi-challenge secure UE.

to trust any banknote to be honest. However, even if such an assumption is made, this is still not a solution: Even perfectly orthogonal states will be deemed to be equal with probability $1/2$ by the SWAP test, and parallel amplification will not work either since the adversary needs to only track a single instance to track the amplified banknote. To fully prevent adversarial tracking attacks that involve tampering with banknotes, one may aim to construct a quantum coin scheme in which the verification procedure consists solely of a projection onto the correct state. This would indeed ensure perfect anonymity even against malicious users. We remark that such a construction is known in the private-key setting [JLS18], but achieving it in the public-key setting appears challenging. Our construction of quantum coins can be viewed as a first step toward this goal. In particular, since our equal-superposition compiler works with arbitrary public states that have delocalized supports, it might be possible to come up with some state that could give us projective verification.

Comparison with [PRV24]. We compare our multi-copy UE scheme with that of [PRV24]. The security notion achieved in [PRV24] is weaker in two respects. First, their construction is limited to the *bounded-copy* setting, where the number of copies is restricted to $q = o(n/\log n)$, with n denoting the key length. Second, they only achieve a weaker *oracular* security notion, where the adversary in the second stage is provided with encryption and decryption oracles rather than the decryption key itself. In contrast, our scheme achieves *standard unbounded multi-copy security*, where the number of copies can be an arbitrarily large polynomial, and the entire decryption key is revealed to the adversary in the second stage.

On the other hand, our scheme relies on the existence of iO and OWFs, whereas their construction relies only on OWFs. In addition, their scheme enjoys an extra property: a ciphertext is fully determined by the message and the key. While our scheme can satisfy this property if the encryption and decryption keys are allowed to differ, requiring them to be identical, as in the standard definition, necessitates that the ciphertext also depend on additional encryption randomness (see [Remark 5](#)). It remains an open problem to construct a multi-copy secure UE scheme that satisfies this additional property while using identical encryption and decryption keys.

Comparison with [AMP25]. We compare our multi-copy certified deletion scheme with that of [AMP25]. In the standard security notion of certified deletion, the decryption key is revealed to the adversary once the adversary provides a valid certificate of deletion. In contrast, [AMP25] considers a weaker *oracular* security notion, where the adversary is given access to an oracle that depends on the decryption key instead of receiving the key itself. On the other hand, our scheme achieves the *standard* certified deletion security, where the entire key is revealed. Moreover, our construction naturally extends to the reusable and public-key settings.

1.3 Organization

The remainder of the paper is organized as follows. In [Section 2](#), we review relevant definitions and known results. In [Section 3](#), we present a theorem that gives our main compiler, which upgrades collusion-resistance into multi-copy security. In [Sections 4](#) and [5](#), we develop the technical tools used for the security proof of the additional folklore construction of quantum coins. In [Section 6](#), we present two constructions of quantum coins. The first ([Section 6.1](#)) is obtained by directly applying the compiler developed in [Section 3](#), while the second ([Section 6.2](#)) is the folklore “equal-superposition” construction whose security we establish using the tools developed in [Sections 4](#) and [5](#). In [Section 7](#), we introduce upgradable quantum coins and present its construction. In [Section 8](#), we describe a generic compiler that transforms single-key SDE into collusion-resistant

SDE, which is then compiled into multi-copy secure SDE by the compiler in [Section 3](#). In [Section 9](#), we demonstrate a conversion from collusion-resistant SDE to multi-challenge secure UE, which is then compiled into multi-copy secure UE by the compiler in [Section 3](#).

Acknowledgments

Generative AI tools were used for language editing. Alper Çakan was supported by the following grants of Vipul Goyal: NSF award 1916939, DARPA SIEVE program, a gift from Ripple, a DoE NETL award, a JP Morgan Faculty Fellowship, a PNC center for financial services innovation award, and a Cylab seed funding award.

2 Preliminaries

2.1 Pseudorandom Functions

Definition 1 (Pseudorandom Function (PRF)). *A pseudorandom function (PRF) is a pair of algorithms $\text{PRF} = (\text{PRF.Gen}, \text{F})$, where $\{F_K : \{0, 1\}^{\ell_1} \rightarrow \{0, 1\}^{\ell_2} \mid K \in \{0, 1\}^\lambda\}$ is a family of efficiently computable functions. Here ℓ_1 and ℓ_2 are polynomials in the security parameter λ . The PRF satisfies the following pseudorandomness condition.*

Pseudorandomness *For any QPT distinguisher $\mathcal{A}$, it holds that*

$$\left| \Pr \left[\mathcal{A}^{F_K(\cdot)}(1^\lambda) \rightarrow 1 \right] - \Pr \left[\mathcal{A}^{\mathcal{U}_{\ell_1, \ell_2}(\cdot)}(1^\lambda) \rightarrow 1 \right] \right| \leq \text{negl}(\lambda), \quad (1)$$

where $K \leftarrow \text{PRF.Gen}(1^\lambda)$, and $\mathcal{U}_{\ell_1, \ell_2}$ denotes a truly random function mapping $\{0, 1\}^{\ell_1}$ to $\{0, 1\}^{\ell_2}$.

Definition 2 (Puncturable PRF). *A puncturable PRF (PPRF) is a tuple of algorithms $\text{PPRF} = (\text{PRF.Gen}, \text{F}, \text{Puncture})$ where $\{F_K : \{0, 1\}^{\ell_1} \rightarrow \{0, 1\}^{\ell_2} \mid K \in \{0, 1\}^\lambda\}$ is a PRF family and satisfies the following two conditions. Note that ℓ_1 and ℓ_2 are polynomials of λ .*

Punctured correctness: *For any polynomial-size set $S \subseteq \{0, 1\}^{\ell_1}$ and any $x \in \{0, 1\}^{\ell_1} \setminus S$, it holds that*

$$\Pr \left[F_K(x) = F_{K_{\notin S}}(x) \mid K \leftarrow \text{PRF.Gen}(1^\lambda), K_{\notin S} \leftarrow \text{Puncture}(K, S) \right] = 1. \quad (2)$$

Pseudorandom at punctured point: *For any polynomial-size set $S \subseteq \{0, 1\}^{\ell_1}$ and any QPT distinguisher $\mathcal{A}$, it holds that*

$$\left| \Pr \left[\mathcal{A}(F_{K_{\notin S}}, \{F_K(x_i)\}_{x_i \in S}) \rightarrow 1 \right] - \Pr \left[\mathcal{A}(F_{K_{\notin S}}, (\mathcal{U}_{\ell_2})^{|S|}) \rightarrow 1 \right] \right| \leq \text{negl}(\lambda), \quad (3)$$

where $K \leftarrow \text{PRF.Gen}(1^\lambda)$, $K_{\notin S} \leftarrow \text{Puncture}(K, S)$ and $\mathcal{U}_{\ell_2}$ denotes the uniform distribution over $\{0, 1\}^{\ell_2}$.

If $S = \{x^*\}$ (i.e., puncturing a single point), we simply write $F_{\neq x^*}(\cdot)$ instead of $F_{K_{\notin S}}(\cdot)$ and consider $F_{\neq x^*}$ as a keyed function.

It is easy to see that the Goldwasser-Goldreich-Micali tree-based construction of PRFs (GGM PRF) [GGM86] from OWF yield puncturable PRFs where the size of the punctured key grows polynomially with the size of the set S being punctured [BW13, BGI14, KPTZ13]. Thus, we have:

Theorem 2 ([GGM86, BW13, BGI14, KPTZ13]). *If OWFs exist, then for any polynomials $\ell_1(\lambda)$ and $\ell_2(\lambda)$, there exists a PPRF that maps ℓ_1 -bits to ℓ_2 -bits.*

2.2 Indistinguishability Obfuscation

In this section, we recall indistinguishability obfuscation (iO).

Definition 3 (Indistinguishability Obfuscation). *An indistinguishability obfuscation (iO) scheme for a class of circuits $\mathcal{C} = \{C_\lambda\}_\lambda$ is an efficient algorithm $i\mathcal{O}$ that satisfies the following.*

Correctness. *For all $\lambda \in \mathbb{N}^+$, $C \in \mathcal{C}_\lambda$ and all inputs x to C ,*

$$\Pr[\tilde{C}(x) = C(x) : \tilde{C} \leftarrow i\mathcal{O}(1^\lambda, C)] = 1.$$

Security. *Let $\mathcal{B}$ be any QPT algorithm that outputs two circuits $C_0, C_1 \in \mathcal{C}$ of the same size, along with quantum auxiliary information R_{aux} , such that $\Pr[\forall x \ C_0(x) = C_1(x) : (C_0, C_1, R_{\text{aux}}) \leftarrow \mathcal{B}(1^\lambda)] \geq 1 - \text{negl}(\lambda)$. Then, for any QPT adversary $\mathcal{A}$,*

$$\left| \Pr[\mathcal{A}(i\mathcal{O}(1^\lambda, C_0), R_{\text{aux}}) = 1 : (C_0, C_1, R_{\text{aux}}) \leftarrow \mathcal{B}(1^\lambda)] - \Pr[\mathcal{A}(i\mathcal{O}(1^\lambda, C_1), R_{\text{aux}}) = 1 : (C_0, C_1, R_{\text{aux}}) \leftarrow \mathcal{B}(1^\lambda)] \right| \leq \text{negl}(\lambda).$$

2.3 Functional Encryption

Definition 4 (Functional Encryption). *An FE scheme FE is a tuple of PPT algorithms (Setup, KG, Enc, Dec).*

$\text{Setup}(1^\lambda) \rightarrow (\text{pk}, \text{msk})$: *The setup algorithm takes a security parameter 1^λ and outputs a public key pk and master secret key msk .*

$\text{KG}(\text{msk}, f) \rightarrow \text{fsk}$: *The key generation algorithm KG takes a master secret key msk and a function f , and outputs a functional decryption key fsk .*

$\text{Enc}(\text{pk}, x) \rightarrow \text{ct}$: *The encryption algorithm takes a public key pk and an input x , and outputs a ciphertext ct .*

$\text{Dec}(\text{fsk}, \text{ct}) \rightarrow y$: *The decryption algorithm takes a functional decryption key fsk and a ciphertext ct , and outputs y .*

Correctness: *We require we have that*

$$\Pr \left[\text{Dec}(\text{fsk}, \text{ct}) = f(x) \mid \begin{array}{l} (\text{pk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda), \\ \text{fsk} \leftarrow \text{KG}(\text{msk}, f), \\ \text{ct} \leftarrow \text{Enc}(\text{pk}, x) \end{array} \right] = 1 - \text{negl}(\lambda).$$

Definition 5 (Adaptive Security for PKFE). *We formalize the experiment $\text{Exp}_\mathcal{A}^{\text{ada-ind}}(1^\lambda, \text{coin})$ between an adversary $\mathcal{A}$ and a challenger for PKFE scheme for $\mathcal{X}, \mathcal{Y}$, and $\mathcal{F}$ as follows:*

1. *The challenger runs $(\text{pk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$ and sends pk to $\mathcal{A}$.*
2. *$\mathcal{A}$ sends arbitrary key queries. That is, $\mathcal{A}$ sends function $f_i \in \mathcal{F}$ to the challenger and the challenger responds with $\text{fsk}_i \leftarrow \text{KG}(\text{msk}, f_i)$ for the i -th query f_i .*
3. *At some point, $\mathcal{A}$ sends (x_0, x_1) to the challenger. If $f_i(x_0) = f_i(x_1)$ for all i , the challenger generates a ciphertext $\text{ct}^* \leftarrow \text{Enc}(\text{pk}, x_{\text{coin}})$. The challenger sends ct^* to $\mathcal{A}$.*

4. Again, $\mathcal{A}$ can send function queries f_i such that $f_i(x_0) = f_i(x_1)$.
5. $\mathcal{A}$ outputs a guess coin' for coin .
6. The experiment outputs coin' .

We say that PKFE is adaptively secure if, for any QPT $\mathcal{A}$, it holds that

$$\text{Adv}_{\text{PKFE}, \mathcal{A}}^{\text{ada-ind}}(\lambda) := \left| \Pr \left[\text{Exp}_{\text{PKFE}, \mathcal{A}}^{\text{ada-ind}}(1^\lambda, 0) \rightarrow 1 \right] - \Pr \left[\text{Exp}_{\text{PKFE}, \mathcal{A}}^{\text{ada-ind}}(1^\lambda, 1) \rightarrow 1 \right] \right| \leq \text{negl}(\lambda). \quad (4)$$

If $\mathcal{A}$ can send only q key queries in $\text{Exp}_{\text{PKFE}, \mathcal{A}}^{\text{ada-ind}}$ where q is a bounded polynomial, we say that PKFE is q -bounded adaptively secure.

2.4 Quantum Information.

Definition 6 (Quantum algorithms with classically determined outputs). *We say that a quantum algorithm has classically determined outputs if it takes a classical input and classical randomness and outputs a pure state that is fully determined by its input and randomness.*

Lemma 1 ([Zha19a, Lemma 5 simplified]). *Let H be a random function $H : \{0, 1\}^{p(\lambda)} \rightarrow \{0, 1\}^{q(\lambda)}$ where $k(\cdot), p(\cdot), q(\cdot)$ are polynomials. Consider a quantum algorithm $\mathcal{A}$ that makes $k(\lambda)$ quantum-queries to H and outputs $(x_1, y_1), \dots, (x_{k+1}, y_{k+1})$. Then, $\Pr[y_1 = H(x_1) \wedge \dots \wedge y_{k+1} = H(x_{k+1})] \leq \text{negl}(\lambda)$, where the probability is taken over the random choice of H and the inherent randomness of the quantum algorithm $\mathcal{A}$.*

Lemma 2 ([BZ13a]). *Let A be a quantum algorithm, and let $\Pr[x]$ be the probability that A outputs x . Let A' be another quantum algorithm obtained from A by pausing A at an arbitrary stage of execution, performing a partial measurement that obtains one of k outcomes, and then resuming A . Let $\Pr'[x]$ be the probability A' outputs x . Then $\Pr'[x] \geq \Pr[x]/k$.*

2.4.1 Testing Quantum Adversaries

Definition 7 (Shift Distance). *For two distributions D_0, D_1 , the shift distance with parameter ϵ , denoted by $\Delta_{\text{Shift}}^\epsilon(D_0, D_1)$, is the smallest quantity δ such that for all $x \in \mathbb{R}$:*

$$\Pr[D_0 \leq x] \leq \Pr[D_1 \leq x + \epsilon] + \delta, \quad \Pr[D_0 \geq x] \leq \Pr[D_1 \geq x - \epsilon] + \delta, \quad (5)$$

$$\Pr[D_1 \leq x] \leq \Pr[D_0 \leq x + \epsilon] + \delta, \quad \Pr[D_1 \geq x] \leq \Pr[D_0 \geq x - \epsilon] + \delta. \quad (6)$$

For two real-valued measurements $\mathcal{M}$ and $\mathcal{N}$ over the same quantum system, the shift distance between $\mathcal{M}$ and $\mathcal{N}$ with parameter ϵ is

$$\Delta_{\text{Shift}}^\epsilon(\mathcal{M}, \mathcal{N}) := \sup_{|\psi\rangle} \Delta_{\text{Shift}}^\epsilon(\mathcal{M}(|\psi\rangle), \mathcal{N}(|\psi\rangle)).$$

Definition 8 (Projective Implementation [Zha20]). *Let:*

- $\mathcal{D}$ be a finite set of distributions over an index set $\mathcal{I}$.
- $\mathcal{P} = \{\mathbf{P}_i\}_{i \in \mathcal{I}}$ be a POVM.
- $\mathcal{E} = \{\mathbf{E}_D\}_{D \in \mathcal{D}}$ be a projective measurement with index set $\mathcal{D}$.

We consider the following measurement procedure.

1. Measure under the projective measurement $\mathcal{E}$ and obtain a distribution D .
2. Output a random sample from the distribution D .

We say $\mathcal{E}$ is the projective implementation of $\mathcal{P}$, denoted by $\text{ProjImp}(\mathcal{P})$, if the measurement process above is equivalent to $\mathcal{P}$.

Theorem 3 ([Zha20, Lemma 1]). Any binary outcome POVM $\mathcal{P} = (\mathbf{P}, \mathbf{I} - \mathbf{P})$ has a unique projective implementation $\text{ProjImp}(\mathcal{P})$.

Definition 9 (Mixture of Projective Measurement [Zha20]). Let $D : \mathcal{R} \rightarrow \mathcal{I}$ where $\mathcal{R}$ and $\mathcal{I}$ are some sets. Let $\{(\mathbf{P}_i, \mathbf{Q}_i)\}_{i \in \mathcal{I}}$ be a collection of binary projective measurement. The mixture of projective measurements associated to $\mathcal{R}$, $\mathcal{I}$, D , and $\{(\mathbf{P}_i, \mathbf{Q}_i)\}_{i \in \mathcal{I}}$ is the binary POVM $\mathcal{P}_D = (\mathbf{P}_D, \mathbf{Q}_D)$ defined as follows.

$$\mathbf{P}_D = \sum_{i \in \mathcal{I}} \Pr[i \leftarrow D(R)] \mathbf{P}_i \quad \mathbf{Q}_D = \sum_{i \in \mathcal{I}} \Pr[i \leftarrow D(R)] \mathbf{Q}_i, \quad (7)$$

where R is uniformly distributed in $\mathcal{R}$.

Definition 10 (Threshold Implementation [Zha20, ALL⁺21]). Let

- $\mathcal{P} = (\mathbf{P}, \mathbf{I} - \mathbf{P})$ be a binary POVM
- $\mathcal{E}$ be the projective measurement in the first step of the measurement procedure in [Definition 8](#).
- $t > 0$.

A threshold implementation of $\mathcal{P}$, denoted by $\text{TI}_t(\mathcal{P})$, is the following measurement procedure.

- Apply $\mathcal{E}$ to a quantum state and obtain $(p, 1 - p)$ as an outcome.
- Output 1 if $p \geq t$, and 0 otherwise.

For any quantum state ρ , we denote by $\text{Tr}[\text{TI}_t(\mathcal{P})\rho]$ the probability that the threshold implementation applied to ρ outputs 1 as Coladangelo et al. did [CLLZ21]. This means that whenever $\text{TI}_t(\mathcal{P})$ appears inside a trace Tr , we treat $\text{TI}_t(\mathcal{P})$ as a projection onto the 1 outcome.

Lemma 3 ([ALL⁺21]). Any binary POVM $\mathcal{P} = (\mathbf{P}, \mathbf{I} - \mathbf{P})$ has a threshold implementation $\text{TI}_t(\mathcal{P})$ for any t .

Theorem 4 (Properties of Threshold Implementation). 1. Applying $\text{TI}_\gamma(\mathcal{P}_D)$ on ρ outputs a binary outcome b' and a collapsed program ρ' .

2. If $b' = 1$, ρ' has success probability at least γ with respect to D . Furthermore, applying $\text{TI}_\gamma(\mathcal{P}_D)$ on ρ' always outputs 1.
3. If $b' = 0$, ρ' has success probability less than γ with respect to D . Furthermore, applying $\text{TI}_\gamma(\mathcal{P}_D)$ on ρ' always outputs 0.

Theorem 5 ([Zha20, ALL⁺21]). Let

- $t > 0$
- $\mathcal{P}$ be a collection of projective measurements indexed by some sets

- ρ be an efficiently constructible mixed state
- D_0 and D_1 be two efficiently samplable and computationally indistinguishable distributions over $\mathcal{I}$.

For any inverse polynomial ϵ , there exists a negligible function δ such that

$$\text{Tr}[\text{TI}_{t-\epsilon}(\mathcal{P}_{D_1})\rho] \geq \text{Tr}[\text{TI}_t(\mathcal{P}_{D_0})\rho] - \delta,$$

where $\mathcal{P}_{D_{\text{coin}}}$ is the mixture of projective measurements associated to $\mathcal{P}$, D_{coin} , and $\text{coin} \in \{0, 1\}$.

Lemma 4 ([ALL⁺21]). For any $\epsilon, \delta, t \in (0, 1)$, any collection of projective measurements $\mathcal{P} = \{(\mathbf{P}_i, \mathbf{I} - \mathbf{P}_i)\}_{i \in \mathcal{I}}$ where $\mathcal{I}$ is some index set, and any distribution D over $\mathcal{I}$, there exists a measurement procedure $\text{ATI}_{\mathcal{P}, D, t}^{\epsilon, \delta}$ that satisfies the following.

- $\text{ATI}_{\mathcal{P}, D, t}^{\epsilon, \delta}$ implements a binary outcome measurement.
- For all quantum state ρ ,
 - $\text{Tr}[\text{ATI}_{\mathcal{P}, D, t-\epsilon}^{\epsilon, \delta}\rho] \geq \text{Tr}[\text{TI}_t(\mathcal{P}_D)\rho] - \delta$ and
 - $\text{Tr}[\text{TI}_{t-\epsilon}(\mathcal{P}_D)\rho] \geq \text{Tr}[\text{ATI}_{\mathcal{P}, D, t}^{\epsilon, \delta}\rho] - \delta$.

For simplicity, we denote the probability of the measurement outputting 1 on ρ by $\text{Tr}[\text{ATI}_{\mathcal{P}, D, t}^{\epsilon, \delta}\rho]$.

- For all quantum state ρ , let ρ' be the post-measurement state after applying $\text{ATI}_{\mathcal{P}, D, t}^{\epsilon, \delta}$ on ρ , and obtaining outcome 1. Then, it holds $\text{Tr}[\text{TI}_{t-2\epsilon}(\mathcal{P}_D)\rho'] \geq 1 - 2\delta$.
- The expected running time is $T_{\mathcal{P}, D} \cdot \text{poly}(1/\epsilon, 1/\log \delta)$, where $T_{\mathcal{P}, D}$ is the combined running time of sampling according to D , of mapping i to $(\mathbf{P}_i, \mathbf{I} - \mathbf{P}_i)$, and of implementing the projective measurement $(\mathbf{P}_i, \mathbf{I} - \mathbf{P}_i)$.

We can easily obtain the following corollary from [Theorem 5](#) and [Lemma 4](#).

Corollary 1. Let

- $\gamma > 0$
- $\mathcal{P}$ be a collection of projective measurements indexed by some sets
- ρ be an efficiently constructible mixed state
- D_0 and D_1 be two efficiently samplable and computationally indistinguishable distributions over $\mathcal{I}$.

For any inverse polynomial ϵ , there exists a negligible function δ such that

$$\text{Tr}[\text{ATI}_{\mathcal{P}, D_1, t-3\epsilon}^{\epsilon, \delta}\rho] \geq \text{Tr}[\text{ATI}_{\mathcal{P}, D_0, t}^{\epsilon, \delta}\rho] - 3\delta,$$

where $\mathcal{P}_{D_{\text{coin}}}$ is the mixture of projective measurements associated to $\mathcal{P}$, D_{coin} , and $\text{coin} \in \{0, 1\}$.

2.5 Pseudorandom Quantum States

In this section, we recall pseudorandom (quantum) states (PRS) [JLS18].

Definition 11 (Pseudorandom Quantum States). *A pseudorandom quantum state (PRS) scheme PRS is a family of quantum states $\{|\phi_K\rangle\}_{K \in \{0,1\}^{c(\lambda)}}$ over $m(\lambda)$ qubits along with the following efficient algorithms.*

- $\text{PRS.Setup}(1^\lambda)$: Takes in the unary representation of the security parameter and outputs a key $K \in \{0,1\}^{c(\lambda)}$.
- $\text{PRS.GenState}(K)$: On input $K \in \{0,1\}^{c(\lambda)}$, outputs a quantum state over $m(\lambda)$ qubits.

We require the following.

Correctness. For all $\lambda \in \mathbb{N}^+$ and all $K \in \{0,1\}^{c(\lambda)}$, the algorithm $\text{PRS.GenState}(K)$ outputs $|\phi_K\rangle$.

Security. We require that for any QPT adversary $\mathcal{A}$ and any polynomial $t(\lambda)$,

$$\left| \Pr_{K \leftarrow \text{PRS.Setup}(1^\lambda)} \left[\mathcal{A}(|\phi_K\rangle^{\otimes t(\lambda)}) = 1 \right] - \Pr_{|\psi\rangle \leftarrow \text{Haar}(m(\lambda))} \left[\mathcal{A}(|\psi\rangle^{\otimes t(\lambda)}) = 1 \right] \right| \leq \text{negl}(\lambda),$$

where $\text{Haar}(m(\lambda))$ denotes the Haar measure over $(\mathbb{C}^2)^{\otimes m}$

Theorem 6 ([JLS18]). *Let $m(\cdot)$ be polynomially bounded. If (post-quantum) one-way functions exist, then there exists a (post-quantum) pseudorandom quantum state scheme PRS that outputs $m(\lambda)$ -qubit states.*

Part I

Tools

3 Purification Compiler

In this section, we give a purification compiler that transforms any quantum state query game that involves multiple samples from a mixed quantum state into a game where instead multiple copies of the purified state is used.

First, we introduce some notation and recall the following lemma. Let $\mathcal{H}_n$ denote the Haar random state distribution on n qubits and let S_t denote the set of permutations on $[t]$. For any distinct $x_1, \dots, x_t \in \{0,1\}^n$, define $|\text{type}(x_1, \dots, x_t)\rangle$ to be

$$|\text{type}(x_1, \dots, x_t)\rangle = \frac{1}{t!} \sum_{\pi \in S_t} \bigotimes_{i \in [t]} |x_{\pi(i)}\rangle.$$

Lemma 5 ([AKY25]). *For any $n, t \in \mathbb{N}^+$,*

$$\left| \mathbb{E}_{|\psi\rangle \leftarrow \mathcal{H}_n} (|\psi\rangle\langle\psi|)^{\otimes t} - \mathbb{E}_{\substack{x_1, \dots, x_t \in \{0,1\}^n \\ \text{all distinct}}} (|\text{type}(x_1, \dots, x_t)\rangle\langle\text{type}(x_1, \dots, x_t)|)^{\otimes t} \right| \leq O\left(\frac{t^2}{2^n}\right).$$

Now we show our main theorem.

Theorem 7 (Purifying State Games). *Let GenState be a QPT algorithm that has classically determined outputs (Definition 6) with the randomness length $r(\lambda)$. Let Exp denote a security experiment between an efficient challenger and an adversary, where the outcome of the experiment is 0 or 1. We will write $\text{Exp}^{\mathcal{A}}$ to denote the experiment when the adversary is the QPT algorithm $\mathcal{A}$. In an execution of Exp , there are ℓ state-query stages. In each state-query stage, the challenger receives a number t from the adversary, and runs $\text{GenState}(st)$ t times using independent classical randomness (and the same st), and submits the outputs to the adversary, where st is an (evolving) internal classical state of the challenger.*

Consider the experiment Exp_{pure} with a modified challenger where for each $i \in [\ell]$ the challenger samples a PRS key k_i for λ -qubit outputs and a PRF key K_i for a pseudorandom function F from $\{0, 1\}^\lambda$ to $\{0, 1\}^{r(\lambda)}$, and all the state-queries are modified as follows: During each state-query stage $i \in [\ell]$, instead of calling $\text{GenState}(st)$ t times with independent randomness, the challenger instead outputs t times the state $\sum_x \alpha_{k_i, x} |x\rangle \otimes |\phi_x\rangle$ where $|\phi_x\rangle$ is the state obtained by calling $\text{GenState}(st; F(K_i, x))$ and $|\psi_{k_i}\rangle = \sum_x \alpha_{k_i, x} |x\rangle$ is the state produced by the PRS scheme with the key k_i .

Then, assuming security of the PRS and PRF, for any QPT adversary $\mathcal{A}$ for Exp_{pure} , there is a QPT adversary $\mathcal{A}'$ for Exp such that $|\Pr[\text{Exp}_{\text{pure}}^{\mathcal{A}} = 1] - \Pr[\text{Exp}^{\mathcal{A}'} = 1]| \leq \text{negl}(\lambda)$. Thus, the original security guarantee holds even for the modified experiment.

If the number of states queried by the adversary in each state-query stage is bounded by some fixed polynomial $q(\lambda)$ for all admissible adversaries, then the above result holds unconditionally (i.e. without assuming one-way functions) by replacing the PRS with a statistical $q(\lambda)$ -design and replacing the PRF with a $2q(\lambda)$ -wise independent function.

Proof. Fix a QPT adversary $\mathcal{A}$. Let Hyb_0 denote $\text{Exp}_{\text{pure}}^{\mathcal{A}}$. Define Hyb_1 by modifying Hyb_0 as follows: For each stage $i \in [\ell]$, instead of using $F(K_i, \cdot)$ when generating the output states, instead use a $2 \cdot t$ -wise independent function H_i . We have $\text{Hyb}_0 \approx \text{Hyb}_1$ by the security of PRF and since t -quantum-query algorithms cannot distinguish between a truly random function and a $2 \cdot t$ wise independent functions [Zha12].

Define Hyb_2 by modifying Hyb_1 as follows: For all state-query stages, sample $|\psi'_i\rangle \leftarrow \mathcal{H}_\lambda$ and when computing the output state, use $|\psi'_i\rangle$ instead of the pseudorandom state $|\psi_{k_i}\rangle$. Note that since we are using (multiple copies of) the Haar random sample $|\psi'_i\rangle$, this experiment is inefficient. We have $\text{Hyb}_1 \approx \text{Hyb}_2$ by the security of the PRS scheme.

Define Hyb_3 by modifying Hyb_2 as follows: For each stage $i \in [\ell]$, first the challenger samples $x_1^i, \dots, x_t^i$, each from $\{0, 1\}^\lambda$ subject to all being distinct (otherwise independent). Then, it prepares the state $|\text{type}(x_1, \dots, x_t)\rangle$ and outputs it instead of $(|\psi'_i\rangle)^{\otimes t}$. We have $\text{Hyb}_2 \approx \text{Hyb}_3$ by Lemma 5.

Now define the adversary $\mathcal{A}'$ for Exp where $\mathcal{A}'$ simulates $\mathcal{A}$ except for the following differences. For each stage $i \in [\ell]$, it samples values $x_1^i, \dots, x_t^i$, each from $\{0, 1\}^\lambda$ subject to all being distinct (otherwise independent), and when it receives the states $|\phi_1^i\rangle, \dots, |\phi_t^i\rangle$ from the challenger, it prepares the state

$$|\eta\rangle = \sum_{\pi \in S_t} |x_{\pi(1)}^i\rangle \otimes |\phi_{\pi(1)}^i\rangle \otimes \dots \otimes |x_{\pi(t)}^i\rangle \otimes |\phi_{\pi(t)}^i\rangle$$

and passes it to $\mathcal{A}$. Note that this state can be prepared efficiently. First, note that, as shown by [BKS⁺18], we can efficiently create the state $\sum_{\pi \in S_t} |\pi(1)\rangle \dots |\pi(t)\rangle$ using a quantum version of the Fisher-Yates (FY) algorithm. Once we create the state $\sum_{\pi \in S_t} |\pi(1)\rangle_{P_1} \dots |\pi(t)\rangle_{P_t}$, we can then coherently run a controlled SWAP algorithm to permute the registers carrying $|x_1^i\rangle |\phi_1^i\rangle, \dots, |x_t^i\rangle |\phi_t^i\rangle$. Finally, since $x_1^i, \dots, x_t^i$ are distinct, we can read the corresponding registers to recompute the permutation, and thus undo the registers $P_1, \dots, P_t$. This perfectly creates the state $|\eta\rangle$.

Define Hyb'_1 by modifying $\mathcal{A}'$ playing Exp : For each stage $i \in [\ell]$, the challenger samples a random function H_i and runs $\text{GenState}(st)$ using the randomness $H_i(x_j^i)$ (for $j \in [t]$). We get $\text{Hyb}'_1 \equiv \text{Exp}^{\mathcal{A}'}$. Finally, define Hyb'_2 by modifying Hyb'_1 as follows: Instead of a random function, sample a $2 \cdot t$ -wise independent function H_i . As before, $\text{Hyb}'_1 \approx \text{Hyb}'_2$ follows by [Zha12]. Finally, it is easy to see that $\text{Hyb}'_2 \equiv \text{Hyb}_3$. Combining the above, we get $\text{Exp}_{\text{pure}}^{\mathcal{A}} \approx_{\text{negl}(\lambda)} \text{Exp}^{\mathcal{A}'}$ as desired. $\square$

4 Quantum-State Read-Once Small-Range Distributions

We first recall *small-range distributions*.

Theorem 8 (Small Range Distributions [Zha12]). *There is a universal constant $C_{\text{SRD}} > 1$ such that, for any sets $\mathcal{X}$ and $\mathcal{Y}$, distribution $\mathcal{D}$ on $\mathcal{Y}$, any integer ℓ , and any quantum algorithm $\mathcal{A}$ making k queries to an oracle $H : \mathcal{X} \rightarrow \mathcal{Y}$, the following two cases are indistinguishable, except with probability less than $C_{\text{SRD}} \frac{k^3}{\ell}$*

- $H(x) = y_x$ where $(y_x)_{x \in \mathcal{X}}$ is a list of samples of $\mathcal{D}$ of size $|\mathcal{X}|$.
- $H(x) = \vec{y}_{P(x)}$ where $(y_i)_{i \in [\ell]}$ is a list of samples of $\mathcal{D}$ of size ℓ and P is a random function $P : \mathcal{X} \rightarrow [\ell]$.

Now we introduce a new result showing that an adversary that obtains superpositions of quantum state samples can be simulated with a state whose support consists only of distinct copies of polynomially many quantum state samples.

Theorem 9 (Quantum-State Read-Once Small-Range Distributions). *Let $\mathcal{X}$ be a set, and let $\mathcal{D}$ be a distribution over pure quantum states with efficient descriptions, both implicitly a sequence of such object indexed by a security parameter λ . Let $\mathcal{O}$ be an oracle with input domain $\mathcal{X}$ and output values consisting of samples from $\mathcal{D}$. That is, $\mathcal{O}$ is constructed by sampling independently a state $|\psi_x\rangle \leftarrow \mathcal{D}$ for each $x \in \mathcal{X}$.*

Let $\mathcal{A}$ be an adversary that makes $k(\lambda)$ (where $k(\cdot)$ is a polynomial) non-adaptive queries to $\mathcal{O}$ such that all queries have negligible weight on each $x \in \mathcal{X}$.⁴ Then, for any polynomial $p(\cdot)$, the execution of $\mathcal{A}$ can be simulated with trace distance at most $1/p(\lambda)$ using the state

$$|\phi_0\rangle = \sum_{\substack{x_1, \dots, x_k \in \mathcal{X} \\ P(x_1), \dots, P(x_k) \text{ distinct}}} c \cdot \alpha_{x_1}^{(1)} \dots \alpha_{x_k}^{(k)} |x_1\rangle \left| \tilde{\psi}_{P(x_1)} \right\rangle \dots |x_k\rangle \left| \tilde{\psi}_{P(x_k)} \right\rangle.$$

where $P : \mathcal{X} \rightarrow [\ell(\lambda)]$ is random function, $\ell = C_{\text{OSRD}} \cdot p^2 k^3$ and $C_{\text{OSRD}} > 1$ is a universal constant.

Proof. We will first construct a sequence of indistinguishable hybrids where modify the oracle of $\mathcal{A}$, and finally we will construct a simulator that simulates the final hybrid. We set the constant C_{OSRD} to be $16 \cdot C_{\text{SRD}}$.

Hyb_0 : The original experiment, where the output is the final state output by $\mathcal{A}$.

Hyb_1 : We modify the experiment as follows. We sample a truly random function $P : \mathcal{X} \rightarrow [\ell(\lambda)]$ at the beginning. Further, we sample $|\tilde{\psi}_i\rangle \leftarrow \mathcal{D}$ for $i \in [\ell]$. Then, instead of answering a query of $\mathcal{A}$ to $\mathcal{O}$ as $\sum \alpha_x^{(i)} |x\rangle \rightarrow \sum \alpha_x^{(i)} |x\rangle |\psi_x\rangle$, we now answer it as $\sum \alpha_x^{(i)} |x\rangle \rightarrow \sum \alpha_x^{(i)} |x\rangle |\tilde{\psi}_{P(x)}\rangle$.

⁴An example is an adversary that makes equal superposition queries (where $|\mathcal{X}|$ is superpolynomial).

Hyb₂ : Let $\sum \alpha_x^{(i)} |x\rangle |\psi_x\rangle$ for $i \in [k]$ be the queries of the adversary (induced by its unitary circuit). Now instead of providing the adversary with the corresponding responses, which is

$$|\phi\rangle = \sum_{x_1, \dots, x_k \in \mathcal{X}} \alpha_{x_1}^{(1)} \dots \alpha_{x_k}^{(k)} |x_1\rangle |\tilde{\psi}_{P(x_1)}\rangle \dots |x_k\rangle |\tilde{\psi}_{P(x_k)}\rangle,$$

we instead give the adversary the state

$$|\phi_0\rangle = \sum_{\substack{x_1, \dots, x_k \in \mathcal{X} \\ P(x_1), \dots, P(x_k) \text{ distinct}}} c \cdot \alpha_{x_1}^{(1)} \dots \alpha_{x_k}^{(k)} |x_1\rangle |\tilde{\psi}_{P(x_1)}\rangle \dots |x_k\rangle |\tilde{\psi}_{P(x_k)}\rangle.$$

where $c \in \mathbb{R}^+$ is the normalizing factor.

Now, we show $|\text{Hyb}_0 - \text{Hyb}_1| \leq \frac{1}{2p}$. First, by our assumption, we have that each $|\psi_x\rangle$ is obtained by sampling a classical string s_x from some distribution $\mathcal{D}'$, and applying a public unitary U (e.g. the universal quantum circuit) to s_i . Now consider an experiment **Hyb'₀** where we simulate the queries of $\mathcal{A}$, but to $\mathcal{D}'$, and then apply the unitary U to convert the sample strings s_x into the quantum states $|\psi_x\rangle$, and then run $\mathcal{A}$ on the resulting state. Observe that this corresponds exactly to **Hyb₀**. Also consider an experiment **Hyb'₁** similarly where we sample a random function $P : \mathcal{X} \rightarrow [\ell]$, then simulate the queries of $\mathcal{A}$ to the modified oracle $\mathcal{D}''$ defined as $s_x = \tilde{s}_{P(x)}$, and then convert them to the corresponding states by applying U and finally running $\mathcal{A}$ on the output. Since $\ell = C_{\text{OSRD}} \cdot p^2 k^3 \geq 2C_{\text{SRD}} p k^3$, by **Theorem 8**, we get $|\text{Hyb}_0 - \text{Hyb}_1| \leq \frac{1}{2p}$.

Finally, we show $|\text{Hyb}_1 - \text{Hyb}_2| \leq \frac{1}{2p}$. We will show that for any states $|\tilde{\psi}_1\rangle, \dots, |\tilde{\psi}_\ell\rangle$, we have $\mathbb{E}_P[|\langle \phi | \phi_0 \rangle|^2] \geq 1 - \frac{k^2}{\ell} - \text{negl}(\lambda)$. Since we have that the trace distance between the hybrids is upper bounded by $\sqrt{1-f}$ where f is the fidelity between the inputs to $\mathcal{A}$ in the two hybrids, and since the fidelity between the two inputs are $\mathbb{E}_P[|\langle \phi | \phi_0 \rangle|^2]$, the above will imply that the trace distance between **Hyb₁** and **Hyb₂** is at most $\sqrt{1 - (1 - \frac{2k^2}{\ell})} \leq \frac{1}{2p}$ due to $\ell = C_{\text{OSRD}} p^2 k^3$ and $C_{\text{OSRD}} > 16$.

Fix the states $|\tilde{\psi}_1\rangle, \dots, |\tilde{\psi}_\ell\rangle$, and the map P , and $|\phi^P\rangle, |\phi_0^P\rangle$ will refer to the corresponding states based on this fixing. We define the unnormalized states

$$\begin{aligned} |\zeta_\emptyset^P\rangle &= \sum_{\substack{x_1, \dots, x_k \in \mathcal{X} \\ P(x_1), \dots, P(x_k) \text{ distinct}}} \alpha_{x_1}^{(1)} \dots \alpha_{x_k}^{(k)} |x_1\rangle |\tilde{\psi}_{P(x_1)}\rangle \dots |x_k\rangle |\tilde{\psi}_{P(x_k)}\rangle \\ |\zeta_S^P\rangle &= \sum_{\substack{x_1, \dots, x_k \in \mathcal{X} \\ P(x_{i_1}), \dots, P(x_{i_{k-|S|}}) \text{ distinct} \\ P(x_{j_1}) = \dots = P(x_{j_{|S|}})}} \alpha_{x_1}^{(1)} \dots \alpha_{x_k}^{(k)} |x_1\rangle |\tilde{\psi}_{P(x_1)}\rangle \dots |x_k\rangle |\tilde{\psi}_{P(x_k)}\rangle. \end{aligned}$$

for all $S \subseteq [k]$, $S \neq \emptyset$ where we write $S = \{j_1, \dots, j_k\}$ and $\bar{S} = \{i_1, \dots, i_{k-|S|}\}$. Observe that we have $|\phi_0^P\rangle = \frac{|\zeta_\emptyset^P\rangle}{\|\zeta_\emptyset^P\|}$, and $|\zeta_{S_1}^P\rangle \perp |\zeta_{S_2}^P\rangle$ for all sets $S_1 \neq S_2$ (including empty sets). Define the unnormalized state $|\zeta^P\rangle = \sum_S |\zeta_S^P\rangle$. Then we have $|\phi\rangle = \frac{|\zeta^P\rangle}{\|\zeta^P\|}$.

By above, we have that, $q^P = |\langle \phi^P | \phi_0^P \rangle|^2 = \frac{|\langle \zeta^P | \zeta_\emptyset^P \rangle|^2}{\|\zeta^P\|^2 \|\zeta_\emptyset^P\|^2}$. Also, since $|\zeta^P\rangle = \sum_S |\zeta_S^P\rangle$ and $|\zeta_{S_1}^P\rangle \perp |\zeta_{S_2}^P\rangle$, we get $q = \frac{|\langle \zeta_\emptyset | \zeta_\emptyset^P \rangle|^2}{\|\zeta^P\|^2 \|\zeta_\emptyset^P\|^2} = \frac{\|\zeta_\emptyset^P\|^2}{\|\zeta^P\|^2}$. Finally, again by pairwise orthogonality, we have

$\|\zeta^P\rangle\|^2 = \sum_S \|\zeta_S^P\rangle\|^2$. Thus, $1 - q = \sum_{S \neq \emptyset} \frac{\|\zeta_S^P\rangle\|^2}{\|\zeta^P\rangle\|^2}$. Now define for each $\gamma, \beta \in [k]$ with $\gamma \neq \beta$

$$|\eta_{\gamma,\beta}^P\rangle = \sum_{\substack{x_1, \dots, x_k \in \mathcal{X} \\ P(x_i) \text{ all distinct for } i \neq \gamma, \beta \\ P(x_\gamma) = P(x_\beta)}} \alpha_{x_1}^{(1)} \dots \alpha_{x_k}^{(k)} |x_1\rangle |\tilde{\psi}_{P(x_1)}\rangle \dots |x_k\rangle |\tilde{\psi}_{P(x_k)}\rangle.$$

It is easy to see that $\sum_{S \neq \emptyset} \|\zeta_S^P\rangle\|^2 \leq \sum_{\gamma \neq \beta} \|\eta_{\gamma,\beta}^P\rangle\|^2$ since the latter expression *adds/counts with repetition* (i.e., same collisions are added more than once). Combining these, we get $1 - q^P \leq \sum_{\gamma \neq \beta} \frac{\|\eta_{\gamma,\beta}^P\rangle\|^2}{\|\zeta^P\rangle\|^2}$. This implies $\mathbb{E}_P[|\langle \phi^P | \phi_0^P \rangle|^2] \geq 1 - \mathbb{E}_P\left[\sum_{\gamma \neq \beta} \frac{\|\eta_{\gamma,\beta}^P\rangle\|^2}{\|\zeta^P\rangle\|^2}\right] = 1 - \sum_{\gamma \neq \beta} \mathbb{E}_P\left[\frac{\|\eta_{\gamma,\beta}^P\rangle\|^2}{\|\zeta^P\rangle\|^2}\right]$ by linearity of expectation.

Now, we claim that for any $\gamma \neq \beta$, we have $\mathbb{E}_P\left[\frac{\|\eta_{\gamma,\beta}^P\rangle\|^2}{\|\zeta^P\rangle\|^2}\right] \leq \frac{1}{\ell} + \text{negl}(\lambda)$, which implies $\sum_{\gamma \neq \beta} \mathbb{E}_P\left[\frac{\|\eta_{\gamma,\beta}^P\rangle\|^2}{\|\zeta^P\rangle\|^2}\right] \leq \binom{k}{2} \cdot \left(\frac{1}{\ell} + \text{negl}(\lambda)\right) \leq \frac{k^2}{\ell} + \text{negl}(\lambda)$. To prove our final claim: It is easy to see that since each $\alpha_x^{(\gamma)}$ is negligible for all $\gamma \in [k], x \in \mathcal{X}$, we have that $\mathbb{E}_P\left[\frac{\|\eta_{\gamma,\beta}^P\rangle\|^2}{\|\zeta^P\rangle\|^2}\right] \leq \text{negl}(\lambda) + \max_{x_1 \neq x_2} \Pr_P[P(x_1) = P(x_2)]$, and we also have $\Pr_P[P(x_1) = P(x_2)] = \frac{1}{\ell}$ for $x_1 \neq x_2$. This completes the proof. $\square$

5 Quantum-Query-Secure Deterministic Signature Schemes

We start by recalling Boneh-Zhandry ([BZ13b]) security, also called *plus-one security*, which requires that an adversary that makes k signature quantum queries cannot output valid signatures for $k+1$ distinct messages.

Definition 12 (BZ Security). *Consider the following game between a challenger and an adversary $\mathcal{A}$.*

BZ – GAME $_{\mathcal{A}}(1^\lambda)$

1. The challenger samples $vk, sk \leftarrow \text{SIG.Setup}(1^\lambda)$.
2. For multiple rounds, the adversary queries for a signature by submitting a quantum state $\sum_{m,w} \alpha_{m,w} |m\rangle |w\rangle$, and the challenger responds with $\sum_{m,w} \alpha_{m,w} |m\rangle |w \oplus \text{SIG.Sign}(sk, m)\rangle$ ⁵ which it computes by running the signing function in superposition.
3. After the query phase, the adversary outputs $k+1$ pairs $(m_1, sig_1), \dots, (m_{k+1}, sig_{k+1})$, where k is the number of queries it made.
4. The challenger checks if all m_i are distinct. If not, it outputs 0 and terminates (adversary loses).

⁵Here, for randomized signature schemes, the randomness for SIG.Sign is supplied either by sampling a single random string r per query, and using that for all signatures in the superposition, or by sampling a PRF key per query and using $F(K, m)$ for each signatures in the superposition. In our construction, there will be no per-query randomness, and the signature scheme will be fully deterministic.

5. The challenger verifies each signature as $\text{SIG.Verify}(vk, m_i, sig_i)$. If all pass, the challenger outputs 1, otherwise it outputs 0.

We say that a signature scheme is Boneh-Zhandry (BZ) secure if for any QPT adversary $\mathcal{A}$, we have

$$\Pr[\text{BZ} - \text{GAME}_{\mathcal{A}}(1^\lambda) = 1] \leq \text{negl}(\lambda).$$

We now state our main theorem.

Theorem 10. *Assuming existence of any subexponentially secure collision-resistant hash function, there exists a **deterministic** signature scheme that satisfies BZ security.*

5.1 Construction

Let $n(\lambda)$ denote the message length. Let F, F_2 be PRF schemes, where the input-output length of F will be clear from context, and F_2 maps $n(\lambda)$ bits to λ bits. Let CSIG be a one-time classical-query-secure deterministic signature scheme that is $2^{-n(\lambda)} \cdot \text{negl}(\lambda)$ -secure, such that two times its verification key length is smaller than the length of the messages it can sign. It is easy to see that all of these constructions can be obtained assuming any subexponentially secure collision-resistant hash function.

For a message m , we will write $\text{Bit}_i(m)$ to denote its i -th bit, and we will write $\text{Prefix}_i(m)$ to denote prefix $\text{Bit}_1(m) || \dots || \text{Bit}_i(m)$, with the convention $\text{Prefix}_0(m) = \emptyset$.

SIG.Sign(sk, m)

1. Sample $vk_\emptyset, sk_\emptyset \leftarrow \text{CSIG.Setup}(1^\lambda)$.
2. Sample $K \leftarrow F.\text{Setup}(1^\lambda)$.
3. Sample $K_2 \leftarrow F_2.\text{Setup}(1^\lambda)$.
4. Set $vk = vk_\emptyset$.
5. Set $sk = (sk_\emptyset, K, K_2)$.
6. Output vk, sk .

Before describing our signing algorithm, we introduce some notation. For any strings a of length up to n , we write vk_a^*, sk_a^* to denote the value computed as $vk_a, sk_a = \text{CSIG.Setup}(1^\lambda; F(K, a))$. Similarly, we write $y_m^* = F_2(K_2, m)$ and $sig_m^* = \text{CSIG.Sign}(sk_m, y_m^*, f((x_{m,t}^*)^1))_{t \in [n]}$ for all $m \in \{0, 1\}^n$.

SIG.Sign(sk, m)

1. Parse $(sk_\emptyset, K, K_2) = sk$.
2. For $t \in [n]$, compute $s_t = vk_{\text{Prefix}_{t-1}(m)||0}^* || vk_{\text{Prefix}_{t-1}(m)||1}^* || \text{CSIG.Sign}(sk_{\text{Prefix}_{t-1}(m)}, vk_{\text{Prefix}_{t-1}(m)||0}^* || vk_{\text{Prefix}_{t-1}(m)||1}^*)$.
3. Output $(s_1, \dots, s_n), y_m^*, sig_m^*$.

SIG.Verify(vk, m, sig)

1. Parse $(s_1, \dots, s_n), y, isig = sig$.
2. For $t \in [n]$, parse $pl_t^0 || pl_t^1 || sigpl_t = s_t$ (lengths of components clear from below).
3. Verify $\text{CSIG.Verify}(vk, pl_1^0 || pl_1^1, sigpl_1)$ and then for $t \in [n-1]$, verify $\text{CSIG.Verify}(pl_t^{\text{Bit}_t(m)}, pl_{t+1}^0 || pl_{t+1}^1, sigpl_{t+1})$. Output 0 and terminate if any of the checks fails.
4. Verify $\text{CSIG.Verify}(pl_n^{\text{Bit}_n(m)}, y, isig)$.
5. Output 1 if all checks passed. Otherwise, output 0.

5.2 Proof of Security

We now prove the security of our scheme through a sequence of hybrids.

Hyb₀: The original game $\text{BZ} - \text{GAME}_{\mathcal{A}}(1^\lambda)$.

Hyb₁: In this hybrid, we modify the experiments so that all PRF queries to F, F_2 are replaced with queries to truly random functions H, H_2 sampled during setup.

Hyb_{2,α} **for** $\alpha \in [n]$: Let $(\tilde{s}_{j,t})_{t \in [n]}, \tilde{y}_j, \tilde{isig}_j, \tilde{m}_j$ for $j \in [k+1]$ be the forgeries (signature and message) produced by the adversary. Let $\tilde{pl}_{j,t}^0 || \tilde{pl}_{j,t}^1 || \tilde{sigpl}_{j,t}$ be the parsing (as in SIG.Verify) of $\tilde{s}_{j,t}$ for all $j \in [k+1], t \in [n]$.

We add an additional check at the end of the game. For all $t \in \{1, \dots, \alpha\}$, for all $j \in [k+1]$, we check if $\tilde{pl}_{j,t}^0 || \tilde{pl}_{j,t}^1 = vk_{\text{Prefix}_{t-1}(\tilde{m}_j)}^* || vk_{\text{Prefix}_{t-1}(\tilde{m}_j)}^*$. If any of the checks fails, we output 0 (adversary fails) and terminate.

Hyb₃: At the end of the game, we add the following additional check. For $j \in [k+1]$, we check if $\tilde{y}_j = H_2(\tilde{m}_j)$. If any of the checks fails, we output 0 (adversary fails) and terminate.

Lemma 6. $\text{Hyb}_0 \approx \text{Hyb}_1$

Proof. This follows by the PRF security of F, F_2 . □

Lemma 7. $\text{Hyb}_1 \approx \text{Hyb}_{2,1}$ and $\text{Hyb}_{2,\alpha} \approx \text{Hyb}_{2,\alpha+1}$ for all $\alpha \in [n-1]$.

Proof. We will prove $\text{Hyb}_{2,\alpha} \approx \text{Hyb}_{2,\alpha+1}$, and $\text{Hyb}_1 \approx \text{Hyb}_{2,1}$ follows from essentially the same argument.

Observe that in $\text{Hyb}_{2,\alpha}$, we already know that (conditioned on adversary passing the checks) we have $\tilde{pl}_{j,\alpha}^0 || \tilde{pl}_{j,\alpha}^1 = vk_{\text{Prefix}_{\alpha-1}(\tilde{m}_j)}^* || vk_{\text{Prefix}_{\alpha-1}(\tilde{m}_j)}^*$. Suppose that adversary's output is such that there exists some index $j' \in [k+1]$ such that $\tilde{pl}_{j',\alpha+1}^0 || \tilde{pl}_{j',\alpha+1}^1 \neq vk_{\text{Prefix}_{\alpha}(\tilde{m}_{j'})}^* || vk_{\text{Prefix}_{\alpha}(\tilde{m}_{j'})}^*$. Observe that this constitutes a forgery for the instance of CSIG with the key pair $(vk_{\text{Prefix}_{\alpha}(\tilde{m}_{j'})}^*, sk_{\text{Prefix}_{\alpha}(\tilde{m}_{j'})}^*)$ since:

- We already checked that $\text{CSIG.Verify}(\tilde{pl}_{j',\alpha}^{\text{Bit}_{\alpha}(\tilde{m}_{j'})}, \tilde{pl}_{j',\alpha+1}^0 || \tilde{pl}_{j',\alpha+1}^1, \tilde{sigpl}_{j',\alpha+1}) = 1$,
- We already checked that $\tilde{pl}_{j',\alpha}^{\text{Bit}_{\alpha}(\tilde{m}_{j'})} = vk_{\text{Prefix}_{\alpha}(\tilde{m}_{j'})}^*$,

- Only message signed with $sk_{\text{Prefix}_\alpha(\tilde{m}_{j'})}^*$ in the experiment is $vk_{\text{Prefix}_\alpha(\tilde{m}_{j'})||0}^* || vk_{\text{Prefix}_\alpha(\tilde{m}_{j'})||1}^*$.

Thus, such j' value cannot exist (except with negligible probability); or otherwise we can create an adversary for the one-time unforgeability game $\mathcal{G}$ of CSIG as follows. We place the challenge from the challenger of $\mathcal{G}$ at the node associated with a random string a of length α , and simulate the experiment $\text{Hyb}_{2,\alpha}$. Then, in case we get $\tilde{m}_{j'} = a$ (which happens with probability $2^{-\alpha}$), we get a forgery for $\mathcal{G}$ as argued above. Since CSIG is at least $2^{-\alpha} \cdot \text{negl}(\lambda)$ secure, we conclude that indeed such j' value cannot exist except with negligible probability. This means that (except with negligible probability), it is already the case in $\text{Hyb}_{2,\alpha}$ that if the adversary wins we have $\tilde{p}_{j,\alpha+1}^0 || \tilde{p}_{j,\alpha+1}^1 = vk_{\text{Prefix}_\alpha(\tilde{m}_j)||0}^* || vk_{\text{Prefix}_\alpha(\tilde{m}_j)||1}^*$ for all $j \in [k+1]$. Thus, the newly added checks in $\text{Hyb}_{2,\alpha+1}$ do not change anything, and hence $\text{Hyb}_{2,\alpha} \approx \text{Hyb}_{2,\alpha+1}$. $\square$

Lemma 8. $\text{Hyb}_{2,t} \approx \text{Hyb}_3$.

Proof. This follows from essentially the same argument as the one for $\text{Hyb}_{2,\alpha} \approx \text{Hyb}_{2,\alpha+1}$. $\square$

Lemma 9. $\Pr[\text{Hyb}_3 = 1] \leq \text{negl}(\lambda)$.

Proof. Observe that in this hybrid, the adversary makes k queries to the random function H_2 , and at the end, (if it wins) it produces $k+1$ pairs $(\tilde{m}_t, \tilde{y}_t)$ such that $H_2(\tilde{m}_t) = \tilde{y}_t$. Therefore, we get $\Pr[\text{Hyb}_3 = 1] \leq \text{negl}(\lambda)$ by Lemma 1. $\square$

Combining the above lemmas completes the security proof.

Part II

Quantum Money

6 Quantum Coins

In this section, we give two compilers that transforms any public-key quantum money mini-scheme into a public-key quantum *coin* scheme, that is, a quantum money scheme where all honest banknotes are identical pure states.

We first recall some definitions.

Definition 13 (Quantum Money Mini-Scheme). *A quantum money mini-scheme is a tuple of QPT algorithms $\text{Bank} = (\text{GenBanknote}, \text{Verify})$ defined as follows:*

- $\text{GenBanknote}(1^\lambda)$: *Outputs a serial number sn and a register R containing a banknote.*
- $\text{Verify}(sn, R)$: *Outputs 1 or 0, denoting if the state in the register R is a valid banknote for the serial number sn .*

We require the following.

Correctness

$$\Pr[\text{Verify}(sn, R) = 1] = 1, \tag{8}$$

where $(sn, R) \leftarrow \text{GenBanknote}(1^\lambda)$.

Unforgeability For public-key scheme, we require that for any QPT adversary $\mathcal{A}$, it holds that

$$\Pr \left[\begin{array}{l} (sn, R) \leftarrow \text{GenBanknote}(1^\lambda) \\ b_1 = b_2 = 1 : \quad \begin{array}{l} R_1, R_2 \leftarrow \mathcal{A}(sn, R) \\ b_1 \leftarrow \text{Verify}(sn, R_1) \\ b_2 \leftarrow \text{Verify}(sn, R_2) \end{array} \end{array} \right] \leq \text{negl}(\lambda)$$

For private-key schemes, the requirement is the same except that the adversary does not receive the serial number sn .

We now define quantum money and quantum coin schemes. Since our focus is public-key schemes, we only define the public-key versions, and the private-key versions are defined similarly as in the case of mini-schemes.

Definition 14 (Quantum Money and Quantum Coins). A (public-key) quantum money scheme consists of the following efficient algorithms.

- $\text{Setup}(1^\lambda)$ outputs a public verification key vk and a secret key sk .
- $\text{GenBanknote}(sk)$ outputs a register.
- $\text{Verify}(vk, R)$ outputs 1 or 0, denoting if the input is a valid banknote.

If $\text{GenBanknote}(sk)$ outputs a fixed pure state $|\psi_{sk}\rangle$, then the scheme is called a quantum coin scheme.

We require correctness as before and also require the following security guarantee.

Unforgeability We require that for any polynomial $t(\cdot)$ and for any QPT adversary $\mathcal{A}$

$$\Pr \left[\begin{array}{l} sk \leftarrow \text{Setup}(1^\lambda) \\ \forall i \in [t(\lambda) + 1] b_i = 1 : \quad \begin{array}{l} (R_i) \leftarrow \text{GenBanknote}(sk) \text{ for } i \in [t(\lambda)] \\ (R_i^{adv})_{i \in [t(\lambda)+1]} \leftarrow \mathcal{A}(vk, (R_i)_{i \in [t(\lambda)]}) \\ (b_i) \leftarrow \text{Verify}(vk, R_i^{adv}) \text{ for } i \in [t(\lambda) + 1] \end{array} \end{array} \right] \leq \text{negl}(\lambda)$$

6.1 Pseudorandom State Compiler

Let $\text{MiniBank} = (\text{MiniBank.Gen}, \text{MiniBank.Verify})$ be a public-key mini-scheme. We assume that MiniBank.Gen has classically determined outputs ([Definition 6](#)); namely, its money state is a pure state fully determined by its classical randomness (and the security parameter). This is a very natural requirement satisfied by the existing schemes.

Let SIG be a classical deterministic signature scheme that satisfies unforgeability security with classical queries. Let F a pseudorandom function - its input-output size will be clear from context. Let PRS be a pseudorandom state scheme.

$\text{Bank.Setup}(1^\lambda)$

1. Sample $K \leftarrow F.\text{Setup}(1^\lambda)$.
2. Sample $vk, sgk \leftarrow \text{SIG.Setup}(1^\lambda)$.
3. Sample $k \leftarrow \text{PRS.Setup}(1^\lambda)$
4. Set $sk = (sgk, K, k)$
5. Output vk, sk .

Bank.GenBanknote(sk)

1. Parse $(sgk, K, k) = sk$.
2. Prepare the state $|\$ \rangle = \sum_x \alpha_x |x\rangle |sn_x\rangle |\text{SIG.Sign}(sgk, sn_x)\rangle |\$_x\rangle$ where $sn_x, |\$_x\rangle = \text{MiniBank.Gen}(1^\lambda; r_x)$ and $r_x = F(K, x)$ and $|\psi_k\rangle = \text{PRS.StateGen}(k) = \sum_x \alpha_x |x\rangle$. $|\$ \rangle$ can be prepared efficiently by running $\text{PRS.StateGen}(k)$ and then running $F(K, \cdot)$, MiniBank.Gen and the signing algorithms coherently.
3. Output $|\$ \rangle$.

Bank.Verify(vk, R)

1. Perform the following operations coherently (as in Gentle Measurement Lemma [Aar16]), then rewind.
 1. Measure all but the last register (the mini-banknote register) of R to obtain id, sn, sig, R' .
 2. Verify the signature by $\text{SIG.Verify}(vk, sn, sig)$.
 3. Verify $\text{MiniBank.Verify}(sn, R')$
2. If all verifications passed, output 1. Otherwise, output 0.

Theorem 11. *Bank is a quantum coin scheme that satisfies correctness and unclonability (counterfeiting) security.*

Proof. The correctness is immediate. By Theorem 7, security reduces to the mixed state case where the adversary receives independent samples from the mini-scheme with signed serial numbers, and the security in that case was proven by [AC13]. $\square$

Since the PRS, PRF and signature schemes can be constructed from one-way functions,

Corollary 2. *Assuming the existence of a public-key mini-scheme whose banknote generation algorithm has classically determined outputs and one-way functions, there exists a public-key quantum coin scheme.*

Since a public-key mini-scheme can be constructed from subspace-hiding obfuscation ([Zha19b]), we also get the following corollary.

Corollary 3. *Assuming the existence of subspace-hiding obfuscation and one-way functions, there exists a public-key quantum coin scheme.*

6.2 Equal Superposition Compiler

In this section, we show that the natural scheme that uses the equal superposition of all possible banknotes as the purified banknote gives a secure scheme.

Let $\text{MiniBank} = (\text{MiniBank.Gen}, \text{MiniBank.Verify})$ be a public-key mini-scheme. We assume that MiniBank.Gen has classically determined outputs (Definition 6).

Let SIG be a classical deterministic signature scheme that satisfies BZ-security (Definition 12). Let F a pseudorandom function - its input-output size will be clear from context. Finally, let $\nu(\lambda)$ be a function such that $1/2^{\nu(\lambda)}$ is negligible (i.e., $\nu(\lambda)$ is superlogarithmic).

Bank.Setup(1^λ)

1. Sample $K \leftarrow F.\text{Setup}(1^\lambda)$.
2. Sample $vk, sgk \leftarrow \text{SIG}.\text{Setup}(1^\lambda)$.
3. Set $sk = (sgk, K)$
4. Output vk, sk .

Bank.GenBanknote(sk)

1. Parse $(sgk, K) = sk$.
2. Prepare the state $|\$ \rangle = \sum_{id \in \{0,1\}^{\nu(\lambda)}} |id\rangle |sn_{id}\rangle |\text{SIG}.\text{Sign}(sgk, sn_{id})\rangle |\$_{id}\rangle$ where $sn_{id}, |\$_{id}\rangle = \text{MiniBank}.\text{Gen}(1^\lambda; r_{id})$ and $r_{id} = F(K, id)$.
3. Output $|\$ \rangle$.

Bank.Verify(vk, R)

1. Perform the following operations coherently (as in Gentle Measurement Lemma [Aar16]), then rewind.
 1. Measure all but the last register (the mini-banknote register) of R to obtain id, sn, sig, R' .
 2. Verify the signature by $\text{SIG}.\text{Verify}(vk, sn, sig)$.
 3. Verify $\text{MiniBank}.\text{Verify}(sn, R')$
2. If all verifications passed, output 1. Otherwise, output 0.

Theorem 12. *Bank is a quantum coin scheme that satisfies correctness and unclonability (counterfeiting) security.*

Since the required signature scheme can be constructed from subexponentially secure collision-resistant hash functions (Theorem 10), we obtain the following corollary.

Corollary 4. *Assuming the existence of a public-key mini-scheme and subexponentially secure collision-resistant hash functions, there exists a public-key quantum coin scheme.*

Since a public-key mini-scheme can be constructed from subspace-hiding obfuscation ([Zha19b]), we also get the following corollary.

Corollary 5. *Assuming the existence of subspace-hiding obfuscation and subexponentially secure collision-resistant hash functions, there exists a public-key quantum coin scheme.*

6.2.1 Proof of Security

In this section, we prove [Theorem 12](#). Correctness follows in a straightforward manner from the correctness of the mini-scheme and the other underlying schemes. It is also evident that the scheme always produces identical copies of the pure state $|\$ \rangle$ as banknotes.

We move onto counterfeiting (unclonability) security, which we will prove through a sequence of hybrids (each of which is constructed by modifying the previous one). Suppose for a contradiction that there exists a QPT adversary $\mathcal{A}$ that wins the counterfeiting game with probability $> 1/p(\lambda)$ for infinitely many values of $\lambda > 0$, where $p(\cdot)$ is a polynomial. Let $k(\cdot)$ denote the number of banknotes requested by the adversary⁶.

Hyb₀ : The original game $\text{PKQM} - \text{CF}_{\mathcal{A}}(1^\lambda)$.

Hyb₁ : During setup, we sample a random function H with the same input-output size as F . When answering the banknote generation queries, we now output $|\$'\rangle$ where

$$|\$'\rangle = \sum_{id \in \{0,1\}^{\nu(\lambda)}} |id\rangle |sn_{id}\rangle |\text{SIG.Sign}(sgk, sn_{id})\rangle |\$_{id}\rangle$$

where $sn_{id}, |\$_{id}\rangle = \text{MiniBank.Gen}(1^\lambda; H(id))$.

Hyb₂ : During setup, we sample a truly random function $Q : \{0,1\}^{\nu(\lambda)} \rightarrow [\ell]$ where $\ell = 16C_{\text{OSRD}}p^2k^3$ and C_{OSRD} is the universal constant from [Theorem 9](#). Then, we sample $\tilde{sn}_i, |\tilde{\$}_i\rangle \leftarrow \text{MiniBank.Gen}(1^\lambda)$ for $i \in [\ell]$. Further, instead of providing the k banknotes as $|\psi\rangle = |\$'\rangle^{\otimes k}$ to the adversary, we output $|\phi\rangle$ where

$$|\phi\rangle = \sum_{\substack{id_1, \dots, id_k \in \{0,1\}^{\nu(\lambda)}: \\ Q(id_1), \dots, Q(id_k) \text{ distinct}}} \left(|id_1\rangle_{\text{ID}_1} |\tilde{sn}_{Q(id_1)}\rangle |\text{SIG.Sign}(sgk, \tilde{sn}_{Q(id_1)})\rangle |\tilde{\$}_{Q(id_1)}\rangle \right) \otimes \dots \\ \otimes \left(|id_k\rangle_{\text{ID}_k} |\tilde{sn}_{Q(id_k)}\rangle |\text{SIG.Sign}(sgk, \tilde{sn}_{Q(id_k)})\rangle |\tilde{\$}_{Q(id_k)}\rangle \right)$$

Hyb₃ : We add an extra check at the end of the experiment, in addition to the banknote verification using Bank.Verify . Let $sn_1^*, \dots, sn_{k+1}^*$ be the serial numbers of the forged banknotes⁷. We check if $sn_i^* \in \{\tilde{sn}_1, \dots, \tilde{sn}_\ell\}$ for all $i \in [k+1]$. If any of the checks fail, the adversary loses and experiment outputs 0.

Hyb₄ : We add yet another additional check at the end of the experiment. We check if the serial numbers $sn_1^*, \dots, sn_{k+1}^*$ are distinct, and if not, the adversary loses and experiment outputs 0.

Hyb₅ : We add yet another additional check at the end of the experiment. We sample a random value $i^* \leftarrow [\ell]$ at the beginning of the game. At the end, if $\tilde{sn}_{i^*}$ does not appear at least twice in $sn_1^*, \dots, sn_{k+1}^*$, the adversary loses and experiment outputs 0.

⁶Note that this does **not** make the construction *bounded collusion*: This k value is only used in the reduction, not in the construction.

⁷More formally, let these be the serial numbers obtained from measuring the serial number registers during verification

Hyb₆ : Instead of giving the adversary the state $|\phi\rangle$, instead give it the state $|\phi'\rangle$ constructed as follows. First add an additional auxiliary register whose space is $\mathbb{C}^{\mathbb{Z}_{k+1}}$ to the end of the state $|\phi\rangle$, and initialize it to $|0\rangle$. Then, for each $j \in 1, \dots, k$, apply a controlled add- j gat from the register ID_j to the auxiliary register with the control bit being 1 if the value id in ID_j is such that $Q(id) = i^*$. Then, give the original registers (i.e. the registers except the measured auxiliary register) to the adversary.

We now show hybrid indistinguishability results. Throughout the remainder of the section, when needed, security reductions will implicitly use PRFs to simulate any random functions (such as Q) involved in the experiments to make them efficient.

Lemma 10. $\text{Hyb}_0 \approx \text{Hyb}_1$

Proof. We simply replaced the PRF queries with a truly random function. The result follows by the PRF security of F . $\square$

Lemma 11. $|\Pr[\text{Hyb}_1 = 1] - \Pr[\text{Hyb}_2 = 1]| < \frac{1}{4p}$.

Proof. Observe that here we are simply using the simulator from the quantum-state read-once small-range distribution lemma ([Theorem 9](#)). $\square$

Lemma 12. $\text{Hyb}_2 \approx \text{Hyb}_3$.

Proof. Observe to simulate these experiments, we only need the signatures $\text{SIG.Sign}(sgk, \tilde{sn}_i)$ for $i \in [\ell]$. By EUF – CMA security of SIG, the result follows. $\square$

Lemma 13. $\text{Hyb}_3 \approx \text{Hyb}_4$.

Proof. Due to BZ security of SIG, we know that (except with negligible probability) the adversary cannot output signatures on $k + 1$ distinct messages. Thus the result follows. $\square$

Lemma 14. $\Pr[\text{Hyb}_5 = 1] \geq \frac{\Pr[\text{Hyb}_4=1]}{\ell}$.

Proof. We know that there is a repeating value among $sn_1^*, \dots, sn_{k+1}^*$, and all these values are in the set $\{\tilde{sn}_1, \dots, \tilde{sn}_\ell\}$. Thus, with probability $1/\ell$, the index i^* we sampled randomly will be the repeating one. $\square$

Lemma 15. $\Pr[\text{Hyb}_6 = 1] \geq \frac{\Pr[\text{Hyb}_5=1]}{k+1}$.

Proof. Since there are $k + 1$ measurement outcomes, the result follows by [Lemma 2](#). $\square$

Lemma 16. $\Pr[\text{Hyb}_6 = 1] \leq \text{negl}(\lambda)$.

Proof. First, note that due to the distinctness of these values in the support of the state $|\phi\rangle$, when we are constructing $|\phi'\rangle$, each state in the support of $|\phi\rangle$ will have its auxiliary register modified at most once, and the value in the auxiliary state will exactly show the index at which the state $|\tilde{\$}_{i^*}\rangle$ is used. Thus, once we measure the auxiliary register, no matter what the measurement outcome is, the resulting state either does not involve at all or uses $|\tilde{\$}_{i^*}\rangle$ exactly once (at the the index equal to the auxiliary register measurement outcome). Thus, we can simulate $|\phi'\rangle$ using one copy of $|\tilde{\$}_{i^*}\rangle$.

Using this, we obtain a reduction to the security of the mini scheme, by construct the following adversary $\mathcal{A}'$ for its security game. $\mathcal{A}'$ simulates $\mathcal{A}$ by sampling independent instance of the mini

scheme on its own for all $i \neq i^*$ as $\tilde{s}n_i, |\tilde{\$}_i\rangle \leftarrow \text{MiniBank.Gen}(1^\lambda)$, and setting $\tilde{s}n_{i^*}, |\tilde{\$}_{i^*}\rangle$ to be the challenge banknote received from its own challenger. At the end, we know that if $\mathcal{A}$ wins, it means that it has output two banknotes that pass verification with the serial number $\tilde{s}n_{i^*}$. Thus, $\mathcal{A}'$ wins the mini scheme security game with probability $\Pr[\text{Hyb}_5 = 1]$. By security of the mini-scheme, we get $\Pr[\text{Hyb}_5 = 1] \leq \text{negl}(\lambda)$. $\square$

Combining all but the last lemma gives us $\Pr[\text{Hyb}_6 = 1] \geq \frac{1}{4p\ell(k+1)} \geq \frac{1}{64C_{\text{OSRD}}p^2k^3 \cdot (k+1)}$, whereas the last lemma says $\Pr[\text{Hyb}_6 = 1] \leq \text{negl}(\lambda)$. Thus, we arrive at a contradiction, which completes the proof of security.

7 Upgradable Quantum Coins

In this section, we introduce the notion of upgradable quantum coins. Then, we give a construction that transforms any public-key mini-scheme into an upgradable quantum coin.

We first introduce the relevant models. We will assume that for quantum money mini-schemes, the banknote generation proceeds in two steps: `MiniBank.Gen` outputs a private string s (e.g. the classical description of a quantum state), and `MiniBank.State`(s) outputs the related banknote state and `MiniBank.SerialNumber`(s) outputs the related serial number. This is a natural model and indeed applies to all existing constructions. Note that in the private-key setting, we can take the serial number to be the same as s .

Definition 15 (Upgradable Quantum Money Mini-Scheme). *Let $\mathfrak{S}_1, \mathfrak{S}_2$ be sets of cryptographic constructions, e.g., $\mathfrak{S}_1 = \{\text{one-way-function}\}$ and $\mathfrak{S}_2 = \{\text{one-way-function, subspace-hiding-obfuscation}\}$. A quantum money mini-scheme is said satisfy upgradable correctness and security with respect to $\mathfrak{S}_1, \mathfrak{S}_2$ if it satisfies the following.*

- *MiniBank.Gen and MiniBank.State only uses $\mathfrak{S}_1$, and private-key security (i.e., unclonability without serial number revealed) is satisfied only assuming $\mathfrak{S}_1$.*
- *MiniBank.SerialNumber(s) uses constructions from $\mathfrak{S}_2$, and the mini-scheme satisfies public-key security (i.e., unclonability with serial number given to adversary) assuming $\mathfrak{S}_2$.*

An example is the subspace-based quantum money construction of [AC13, Zha19b]. It satisfies upgradable security with $\mathfrak{S}_1 = \{\text{one-way-function}\}$ and $\mathfrak{S}_2 = \{\text{subspace-hiding-obfuscation}\}$ where `MiniBank.SerialNumber` would be the subspace-hiding obfuscation construction.

Definition 16 (Upgradable Quantum Coin). *Let $\mathfrak{S}_1, \mathfrak{S}_2$ be sets of cryptographic constructions. A quantum coin scheme with upgradable correctness and security with respect to $(\mathfrak{S}_1, \mathfrak{S}_2)$ is defined to be a quantum coin scheme with an additional algorithm `Bank.Reveal` that satisfies the following.*

- **Before Reveal:** *Banknotes can be generated assuming only $\mathfrak{S}_1$, without using any information about $\mathfrak{S}_2$. No key is published initially. Bank satisfies correctness and counterfeiting (unclonability) security assuming only $\mathfrak{S}_1$. Banknote verification by users is done through comparison-based-verification (see [BS21]).*
- **After Reveal:** *Bank.Reveal takes as input bank's secret key, and a randomized program P (which uses constructions from $\mathfrak{S}_2$), and outputs a classical public verification key vk which is published. The scheme (with vk published) satisfies full-fledged public-key quantum money correctness and security assuming $\mathfrak{S}_2$.*

7.1 Construction

Let **MiniBank** be an upgradable quantum-money mini-scheme. Let $\mathfrak{S}_1, \mathfrak{S}_2$ be sets of cryptographic assumptions. Assume that **MiniBank** is secure assuming $\mathfrak{S}_1$ in the private mode, and it is correct and secure assuming $\mathfrak{S}_2$ in the public mode.

Let **SIG** be a classical deterministic signature scheme that satisfies unforgeability security with classical queries. Let F a pseudorandom function - its input-output size will be clear from context. Let **NPKE** be a non-committing public-key encryption scheme. Let $\nu(\lambda)$ be a function such that $2^{\nu(\lambda)}$ is superpolynomial (i.e., $\nu(\lambda)$ is superlogarithmic). Finally, let **1FE** be a public-key functional encryption scheme that satisfies simulation security for a single functional-key with $2^{-\nu(\lambda)} \cdot \text{negl}(\lambda)$ security.

We move onto our construction below. The construction will be upgraded by publishing **Bank.Reveal**($sk, P = \text{MiniBank.SerialNumber}$) once an implementation of **MiniBank.SerialNumber** with $\mathfrak{S}_2$ becomes available.

Bank.Setup(1^λ)

1. Sample $K \leftarrow F.\text{Setup}(1^\lambda)$.
2. Sample $pk, msk \leftarrow \text{1FE.Setup}(1^\lambda)$.
3. Sample $npk, nsk \leftarrow \text{NPKE.Setup}(1^\lambda)$.
4. Sample $svk, sgk \leftarrow \text{SIG.Setup}(1^\lambda)$.
5. Sample $k \leftarrow \text{PRS.Setup}(1^\lambda)$.
6. Set $sk = (K, pk, msk, npk, nsk, svk, sgk, k)$.
7. Output sk .

Bank.GenBanknote(sk)

1. Parse $(K, pk, msk, npk, nsk, svk, sgk, k) = sk$.
2. Prepare the state $|\$ \rangle = \sum_{id \in \{0,1\}^{\nu(\lambda)}} \alpha_{id} |id\rangle |\text{NPKE.Enc}(npk, \text{1FE.Enc}(pk, s_{id} || r_{id}^4 || sgk; r_{id}^2); r_{id}^3)\rangle |\$_{id}\rangle$
 where $r_{id}^1 || r_{id}^2 || r_{id}^3 || r_{id}^4 = F(K, id)$, $s_{id} = \text{MiniBank.Gen}(1^\lambda; r_{id}^1)$ and $|\$_{id}\rangle = \text{MiniBank.State}(s_{id})$
 and $|\psi_k\rangle = \text{PRS.StateGen}(k) = \sum_{id} \alpha_{id} |id\rangle$.
3. Output $|\$ \rangle$.

Bank.Reveal(sk, P)

1. Parse $(K, pk, msk, npk, nsk, svk, sgk) = sk$.
2. Let f_P be the circuit that takes as input $s || r || k$ and outputs $\text{SIG.Sign}(k, P(s; r)) || P(s; r)$.
3. Sample $fsk \leftarrow \text{1FE.KeyGen}(msk, f_P)$.
4. Output $vk = (nsk, svk, fsk)$.

Bank.Verify(vk, R)

1. Parse $(nsk, svk, fsk) = vk$.
2. Measure all but the last register (the mini-banknote register) of R to obtain id, ct, R' .
3. Decrypt $ct' \leftarrow \text{NPKE.Dec}(nsk, ct)$.
4. Decrypt $sig || sn \leftarrow \text{1FE.Dec}(fsk, ct')$.
5. Verify the signature by $\text{SIG.Verify}(svk, sn, sig)$; output 0 and terminate if verification fails.
6. Execute the verification code inside sn on R' , output its output.

Since the required PRF (by [Zha12]), PRS, non-committing encryption (by [HMNY21]) and functional encryption schemes (by [SS10, GVW12]) can be constructed from PKE with subexponential security, we get the following result.

Theorem 13. *Bank is an upgradable quantum coin scheme that satisfies correctness and upgradable security with respect to $(\mathfrak{S}_1 \cup \{\text{subexp} - \text{PKE}\}, \mathfrak{S}_2 \cup \{\text{subexp} - \text{PKE}\})$.*

Since we can instantiate the mini-scheme with $\mathfrak{S}_1 = \{\text{one-way-function}\}$ and $\mathfrak{S}_2 = \{\text{subspace-hiding} - \text{obfuscation}\}$ by [Zha19b], we get the following corollary.

Corollary 6. *There exists an upgradable quantum coin scheme that satisfies correctness and upgradable security with respect to $\mathfrak{S}_1 = \{\text{subexp} - \text{PKE}\}$ and $\mathfrak{S}_2 = \{\text{subexp} - \text{PKE}, \text{subspace} - \text{hiding} - \text{obfuscation}\}$.*

7.2 Proof of Security

Proof of security before the reveal phase simply follows by Theorem 7. We will prove the counterfeiting security of the scheme after the reveal phase, that is, once $vk = (nsk, svk, fsk)$ is public.

Hyb₀ : The original counterfeiting security game.

Hyb₁ : When answering the banknote queries, instead of computing $r_{id}^1 || r_{id}^2 || r_{id}^3 || r_{id}^4 = F(K, id)$, we now compute $r_{id}^1 || r_{id}^2 || r_{id}^3 || r_{id}^4 = H(id)$, where H is a truly random function sampled at setup.

Let $\mathcal{S}$ be a simulator, promised by the simulation-security of 1FE, such that $(pk, fsk, \text{1FE.Enc}(pk, pl)) \approx_{2^{-\nu(\lambda)} \cdot \text{negl}(\lambda)} (pk, fsk, \mathcal{S}(pk, fsk, f_P(pl)))$.

Hyb_{2,j} **for** $j \in \{0, 1\}^{v(\lambda)}$:

When answering the banknote queries, we now output

$$|\$^{(j)}\rangle = \sum_{id \in \{0,1\}^{\nu(\lambda)}} |id\rangle |a_{id}\rangle |\$_{id}\rangle$$

where we define $a_{id} = \text{NPKE.Enc}(npk, \text{1FE.Enc}(pk, s_{id} || r_{id}^4 || sgk; r_{id}^2); r_{id}^3)$ for $id > j$ and define

$$a_{id} = \text{NPKE.Enc}(S(pk, fsk, \text{SIG.Sign}(sgk, P(s_{id}; r_{id}^4)) || P(s_{id}; r_{id}^4); H'(id)); r_{id}^3)$$

for $id \leq j$.

Lemma 17. $\text{Hyb}_0 \approx \text{Hyb}_1$

Proof. We simply replaced the PRF queries with a truly random function, the result follows by the PRF security. $\square$

Lemma 18. $\text{Hyb}_1 \approx_{2^{-\nu(\lambda)} \cdot \text{negl}(\lambda)} \text{Hyb}_{2,0}$ and $\text{Hyb}_{2,j} \approx_{2^{-\nu(\lambda)} \cdot \text{negl}(\lambda)} \text{Hyb}_{2,j+1}$ for all $j \in \{0, 1\}^{\nu(\lambda)}$.⁸

Proof. The only difference between the two neighboring hybrids is switching from the real ciphertext to the simulated ciphertext for $id = j + 1$. The result thus follows by the simulation security of 1FE. $\square$

Corollary 7. $\text{Hyb}_0 \approx \text{Hyb}_{2,2^{\nu(\lambda)}}$.

Proof. This follows by combining the above lemmas. $\square$

Lemma 19. $\Pr[\text{Hyb}_{2,2^{\nu(\lambda)}} = 1] = \text{negl}(\lambda)$.

Proof. In this hybrid, we answer the banknote queries with the state

$$|\phi\rangle = \sum_{id} \alpha_{id} |id\rangle |\text{NPKE.Enc}(S(pk, fsk, \text{SIG.Sign}(sgk, P(s_{id}; r_{id}^4)) || P(s_{id}; r_{id}^4); H'(id)); r_{id}^3)) | \$_{id}\rangle$$

Observe that this state can be simulated using the state

$$|\phi\rangle = \sum_{id} \alpha_{id} |id\rangle |P(s_{id}; r_{id}^4)\rangle |\text{SIG.Sign}(sgk, P(s_{id}; r_{id}^4))\rangle | \$_{id}\rangle$$

by sampling the instances of 1FE and NPKE ourselves. Finally, observe that $|\phi\rangle$ is exactly the banknote states that the compiler in [Section 6](#). Since the compiled scheme's security is proven in [Section 6](#) assuming the security of the mini scheme and SIG, it follows that $\Pr[\text{Hyb}_{2,2^{\nu(\lambda)}} = 1] = \text{negl}(\lambda)$ assuming the security of SIG and the constructions $\mathfrak{S}_2$ $\square$

By above we conclude $\Pr[\text{Hyb}_0 = 1] = \text{negl}(\lambda)$, which completes the security proof.

Part III

Multi-Copy Unclonable Schemes

8 Single-Decryptor Encryption

In this section, we construct multi-copy secure single-decryptor encryption (SDE), where decryption keys are copy-protected.

In [Section 8.1](#), we recall the relevant definitions of collusion-resistant SDE. In [Section 8.2](#), we present a generic compiler that transforms a single-key secure SDE scheme into a collusion-resistant one. In [Section 8.3](#), we provide a security proof for this compiler. In [Section 8.4](#), we show that the security notion proven in [Section 8.3](#) also implies search-based security in the identical-challenge setting, where all parties receive the same challenge ciphertext. Looking ahead, the identical-challenge security plays a crucial role in the conversion to unclonable encryption presented in [Section 9](#). Finally, in [Section 8.5](#), we apply [Theorem 7](#) to obtain a multi-copy secure scheme.

⁸Here, the strings are associated with the numbers $0, 1, \dots, 2^{\nu(\lambda)}$ in the natural way.

8.1 Definitions of Collusion-Resistant SDE

In this section, we provide some definitions related to *single-decryptor encryption* (i.e. copy-protected decryption keys). Unless otherwise specified, we will refer to the security notions where the adversary receives any (unbounded) polynomial number of keys and tries to produce one-more decryption adversary.

Definition 17 (Single-Decryptor Encryption (Syntax)). *An SDE scheme SDE is a tuple of QPT algorithms (Setup, KG, Enc, Dec) with plaintext space $\mathcal{M}$.*

$\text{Setup}(1^\lambda) \rightarrow (\text{pk}, \text{msk})$: *The setup algorithm takes as input the security parameter 1^λ and outputs a public encryption key pk and a master secret key msk .*

$\text{KG}(\text{msk}) \rightarrow (\text{pk}, \text{sk})$: *The key generation algorithm takes as input msk and outputs a quantum decryption key sk .*

$\text{Enc}(\text{pk}, m) \rightarrow \text{ct}$: *The encryption algorithm takes as input pk and a plaintext $m \in \mathcal{M}$ and outputs a ciphertext ct .*

$\text{Dec}(\text{sk}, \text{ct}) \rightarrow m'$: *The decryption algorithm takes as input sk and ct and outputs a plaintext $m' \in \mathcal{M}$ or $\perp$.*

Decryption correctness: *There exists a negligible function negl such that for any $m \in \mathcal{M}$,*

$$\Pr \left[\text{Dec}(\text{sk}, \text{ct}) = m \mid \begin{array}{l} (\text{pk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda) \\ \text{sk} \leftarrow \text{KG}(\text{msk}) \\ \text{ct} \leftarrow \text{Enc}(\text{pk}, m) \end{array} \right] = 1 - \text{negl}(\lambda). \quad (9)$$

For defining security of SDE, we introduce the notions of quantum decryptors following [CLLZ21].

Definition 18 (Quantum Decryptor [CLLZ21]). *Let δ and U be a quantum state and a general quantum circuit acting on $n + m$ qubits, where m is the number of qubits of δ and n is the length of ciphertexts. A quantum decryptor for ciphertexts is a pair (δ, U) .*

When we say that we run the quantum decryptor (δ, U) on ciphertext ct , we execute the circuit U on inputs $|\text{ct}\rangle \otimes \delta$.

Definition 19 (Testing a (CPA-Style) Quantum Decryptor [CLLZ21]). *Let $\gamma \in [0, 1]$. Let pk and (m_0, m_1) be a public key and a pair of messages, respectively. A test for a γ -good quantum decryptor with respect to pk and (m_0, m_1) is the following procedure.*

- *The procedure takes as input a quantum decryptor $\text{qD} = (\delta, U)$.*
- *Let $\mathcal{P} = (\mathbf{P}, \mathbf{I} - \mathbf{P})$ be the following mixture of projective measurements acting on a quantum state δ' :*
 - *Sample a uniformly random coin $\text{coin} \leftarrow \{0, 1\}$ and compute $\text{ct} \leftarrow \text{Enc}(\text{pk}, \text{m}_{\text{coin}})$.*
 - *Run $\text{coin}' \leftarrow \text{qD}'(\text{ct})$, where $\text{qD}' = (\delta', U)$. If $\text{coin}' = \text{coin}$, output 1, otherwise output 0.*

Let $\text{TI}_{1/2+\gamma}(\mathcal{P})$ be the threshold implementation of $\mathcal{P}$ with threshold $\frac{1}{2} + \gamma$. Apply $\text{TI}_{1/2+\gamma}(\mathcal{P})$ to δ . If the outcome is 1, we say that the test passed, otherwise the test failed.

We say that a quantum decryptor $\text{qD} = (\delta, U)$ is a γ -good quantum decryptor with respect to pk and (m_0, m_1) if it passes the above test with probability 1.

Definition 20 (Testing a Search-Type Quantum Decryptor [CLLZ21]). *Let $\gamma \in [0, 1]$. Let $\mathbf{pk}$ be a public key. A test for a γ -good search-type quantum decryptor with respect to $\mathbf{pk}$ is the following procedure.*

- *The procedure takes as input a quantum decryptor $\mathbf{qD} = (\delta, \mathbf{U})$.*
- *Let $\mathcal{P} = (\mathbf{P}, \mathbf{I} - \mathbf{P})$ be the following mixture of projective measurements acting on a quantum state δ' :*
 - *Sample a uniformly random $\mathbf{m} \leftarrow \mathcal{M}$ and compute $\mathbf{ct} \leftarrow \text{Enc}(\mathbf{pk}, \mathbf{m})$.*
 - *Run $\mathbf{m}' \leftarrow \mathbf{qD}'(\mathbf{ct})$, where $\mathbf{qD}' = (\delta', \mathbf{U})$. If $\mathbf{m}' = \mathbf{m}$, output 1, otherwise output 0.*

Let $\text{TI}_{1/|\mathcal{M}|+\gamma}(\mathcal{P})$ be the threshold implementation of $\mathcal{P}$ with threshold $\frac{1}{|\mathcal{M}|} + \gamma$. Apply $\text{TI}_{1/|\mathcal{M}|+\gamma}(\mathcal{P})$ to δ . If the outcome is 1, we say that the test passed, otherwise the test failed.

We say that a quantum decryptor $\mathbf{qD} = (\delta, \mathbf{U})$ is a γ -good search-type quantum decryptor with respect to $\mathbf{pk}$ if it passes the above test with probability 1.

Now we are ready to state the security notions of SDE. Specifically, we define two notions: collusion-resistant strong anti-piracy and collusion-resistant strong search anti-piracy, following [CLLZ21, CG24]. Intuitively, the former corresponds to indistinguishability-based security, while the latter corresponds to search-based security. Unlike in standard public-key encryption, no relationship between these two notions is currently known.⁹

Definition 21 (Collusion-Resistant Strong Anti-Piracy). *Let $\gamma \in [0, 1]$. Let $\text{SDE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$ be a public-key SDE scheme. We consider the collusion-resistant strong anti-piracy game $\text{Exp}_{\text{SDE}, \mathcal{A}}^{\text{strong-anti-piracy}}(\lambda, \gamma)$ between the challenger and an adversary $\mathcal{A}$ below.*

1. *The challenger generates $(\mathbf{pk}, \mathbf{msk}) \leftarrow \text{Setup}(1^\lambda)$ and sends $(1^\lambda, \mathbf{pk})$ to $\mathcal{A}$.*
2. *$\mathcal{A}$ sends 1^q to the challenger, then the challenger generates $\mathbf{sk}_i \leftarrow \text{KG}(\mathbf{msk})$ for $i \in [q]$, and sends $(\mathbf{sk}_1, \dots, \mathbf{sk}_q)$ to $\mathcal{A}$.*
3. *$\mathcal{A}$ outputs $\{(\mathbf{m}_{i,0}, \mathbf{m}_{i,1})\}_{i \in [q+1]} \in \mathcal{M}^{2(q+1)}$ and $(q+1)$ (possibly entangled) quantum decryptors $\mathbf{qD}_1 = (\delta[\mathbf{R}_1], \mathbf{U}_1), \dots, \mathbf{qD}_{q+1} = (\delta[\mathbf{R}_{q+1}], \mathbf{U}_{q+1})$, where $|\mathbf{m}_{i,0}| = |\mathbf{m}_{i,1}|$ for all $i \in [q+1]$, δ is a quantum state over registers $\mathbf{R}_1, \dots, \mathbf{R}_{q+1}$, and each $\mathbf{U}_i$ is a unitary quantum circuit.*
4. *The challenger runs the tests for a γ -good quantum decryptor with respect to $\mathbf{pk}$ and $(\mathbf{m}_{i,0}, \mathbf{m}_{i,1})$ on $\mathbf{qD}_i$ for all $i \in [q+1]$. The challenger outputs 1 if all tests pass; otherwise outputs 0.*

We say that SDE is collusion-resistant strong γ -anti-piracy secure if for any QPT adversary $\mathcal{A}$, it satisfies that

$$\text{Adv}_{\text{SDE}, \mathcal{A}}^{\text{strong-anti-piracy}}(\lambda, \gamma) := \Pr \left[\text{Exp}_{\text{SDE}, \mathcal{A}}^{\text{strong-anti-piracy}}(\lambda, \gamma) = 1 \right] = \text{negl}(\lambda). \quad (10)$$

We say that SDE is collusion-resistant strong anti-piracy secure if it is strong γ -anti-piracy secure for any inverse polynomial γ .

Definition 22 (Collusion-Resistant Strong Search Anti-Piracy). *Let $\gamma \in [0, 1]$. Let $\text{SDE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$ be a public-key SDE scheme. We consider the strong-search anti-piracy game $\text{Exp}_{\text{SDE}, \mathcal{A}}^{\text{strong-search-anti-piracy}}(\lambda, \gamma)$ between the challenger and an adversary $\mathcal{A}$ below.*

⁹A recent work [KY25] showed that collusion-resistant strong anti-piracy implies collusion-resistant strong search anti-piracy in the single-key setting, but their proof does not extend to the collusion-resistant setting.

1. The challenger generates $(\text{pk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$ and sends $(1^\lambda, \text{pk})$ to $\mathcal{A}$.
2. $\mathcal{A}$ sends 1^q to the challenger, then the challenger generates $\text{sk}_i \leftarrow \text{KG}(\text{msk})$ for $i \in [q]$, and sends $(\text{sk}_1, \dots, \text{sk}_q)$ to $\mathcal{A}$.
3. $\mathcal{A}$ outputs $(q+1)$ (possibly entangled) quantum decryptors $\mathbf{qD}_1 = (\delta[R_1], U_1), \dots, \mathbf{qD}_{q+1} = (\delta[R_{q+1}], U_{q+1})$, where δ is a quantum state over registers $R_1, \dots, R_{q+1}$, and each U_i is a unitary quantum circuit.
4. The challenger runs the tests for a γ -good search-type quantum decryptor with respect to pk on $\mathbf{qD}_i$ for all $i \in [q+1]$. The challenger outputs 1 if all tests pass; otherwise outputs 0.

We say that SDE is strong-search γ -anti-piracy secure if for any QPT adversary $\mathcal{A}$, it satisfies that

$$\text{Adv}_{\text{SDE}, \mathcal{A}}^{\text{strong-search-anti-piracy}}(\lambda, \gamma) := \Pr \left[\text{Exp}_{\text{SDE}, \mathcal{A}}^{\text{strong-search-anti-piracy}}(\lambda, \gamma) = 1 \right] = \text{negl}(\lambda). \quad (11)$$

We say that SDE is collusion-resistant strong-search anti-piracy secure if it is strong-search γ -anti-piracy secure for any inverse polynomial γ .

We also define the single-key version of strong search anti-piracy.

Definition 23 (Single-Key Strong Anti-Piracy). We say that an SDE scheme satisfies single-key strong anti-piracy security if it satisfies the requirement of [Definition 21](#) when q is limited to 1.

We recall the following theorem about the existence of single-key SDE.

Theorem 14 ([KY25]). If there exist a secure iO for polynomial size classical circuits and OWFs, then there exists an SDE scheme that satisfies single-key strong anti-piracy security with message space $\{0, 1\}^{\ell_m(\lambda)}$ for any polynomial ℓ_m . Moreover, its key generation algorithm has classically determined outputs ([Definition 6](#)).

8.2 Generic Compiler from Single-Key to Collusion-Resistant SDE

We construct a collusion-resistant single-decryptor encryption scheme $\text{SDE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$ by using the following building blocks.

- Single-key SDE $\text{One} = \text{One}(\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$.
- Adaptively secure PKFE $\text{FE} = \text{FE}(\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$.
- Puncturable PRF $\text{PRF} = (\text{PRF.Gen}, \text{F}, \text{Puncture})$.

We set $\ell_{\text{pk}} := |\text{one.pk}|$ and $\ell_{\text{ct}} := |\text{one.ct}|$, where $|\text{one.pk}|$ and $|\text{one.ct}|$ are the public key and ciphertext length of One, respectively. Let ℓ_m be the plaintext length.

Setup (1^λ) :

- Generate $(\text{fe.pk}, \text{fe.msk}) \leftarrow \text{FE.Setup}(1^\lambda)$.
- Output $(\text{pk}, \text{msk}) := (\text{fe.pk}, \text{fe.msk})$.

KG (msk) :

- Parse $\text{fe.msk} = \text{sk}$ and generate $(\text{one.pk}, \text{one.sk}) \leftarrow \text{One.Setup}(1^\lambda)$.

- Generate $\text{one.sk} \leftarrow \text{One.KG}(\text{one.sk})$.
- Construct a circuit $\text{RE}[\text{one.pk}]$, which is described in [Figure 1](#). Note that we assume that each one.pk is associated with a random tag $\tau_{\text{one.pk}}$ though we do not explicitly write $\tau_{\text{one.pk}}$.
- Generate $\text{fe.fsk} \leftarrow \text{FE.KG}(\text{fe.msk}, \text{RE}[\text{one.pk}])$.
- Output $\text{sk} := (\text{one.sk}, \text{fe.fsk})$.

$\text{Enc}(\text{pk}, \text{m})$:

- Parse $\text{fe.pk} = \text{pk}$.
- Choose $K \leftarrow \text{PRF.Gen}(1^\lambda)$.
- Set $x := \text{m} \| K \| 0 \| 0^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}}$.
- Compute $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, x)$.
- Output $\text{ct} := \text{fe.ct}$.

$\text{Dec}(\text{sk}, \text{ct})$:

- Parse $\text{sk} = (\text{one.sk}, \text{fe.fsk})$ and $\text{ct} = \text{fe.ct}$.
- Compute $\text{one.ct} \leftarrow \text{FE.Dec}(\text{fe.fsk}, \text{fe.ct})$.
- Compute and output $\text{m}' \leftarrow \text{One.Dec}(\text{one.sk}, \text{one.ct})$.

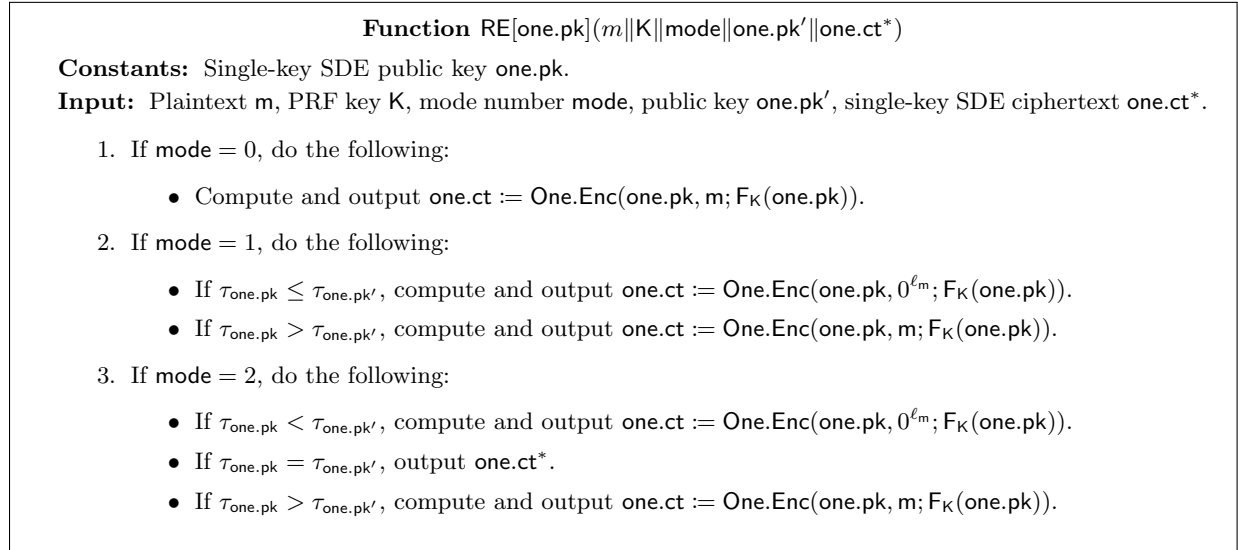


Figure 1: The description of $\text{RE}[\text{one.pk}]$

Theorem 15. *Assume that One is single-key strong anti-piracy secure, FE is adaptively secure, and PRF is a puncturable PRF. Then, SDE satisfies*

- *strong anti-piracy security ([Definition 21](#)),*
- *strong search anti-piracy security ([Definition 22](#)).*

Remark 1. *In the proof of [Theorem 15](#) below, we use PI and TI for simplicity, which are inefficient. In reality, when actually constructing reductions to the security of the underlying primitives, we need to use the efficient versions API and ATI. However, as shown by [\[Zha20, CLLZ21\]](#), by an appropriate choice of parameters, this replacement can be made without affecting the proof.*

8.3 Proof of Theorem 15

We only prove the strong anti-piracy security, and strong search anti-piracy security follows similarly - see Remark 2.

Now we move onto proving strong anti-piracy security. Although our basic proof strategy follows that by Liu et al. [LLQZ22], some details are different. Assume, for the sake of contradiction, that there exist inverse polynomials $\gamma(\cdot)$ and $\nu(\cdot)$, and a polynomial $q(\cdot)$, and an adversary $\mathcal{A}$ such that for infinitely many $\lambda \in \mathbb{N}^+$, $\mathcal{A}$ outputs two messages $(\mathbf{m}_0, \mathbf{m}_1)$ and a state δ over $q+1$ registers satisfying:

$$\text{Tr}[(\text{TI}_{1/2+\gamma} \otimes \text{TI}_{1/2+\gamma} \otimes \cdots \otimes \text{TI}_{1/2+\gamma})\delta] \geq \nu. \quad (12)$$

We now consider the event where all tests $(\text{TI}_{1/2+\gamma} \otimes \text{TI}_{1/2+\gamma} \otimes \cdots \otimes \text{TI}_{1/2+\gamma})$ return outcome 1. Conditioned on this event, let δ^* denote the resulting post-measurement state over the $q+1$ registers. By assumption, this event occurs with probability at least ν . Let $\text{sk}_i := (\text{one.sk}_i, \text{fe.fsk}_i)$ be the i -th quantum decryption key, where $\text{fe.fsk}_i \leftarrow \text{FE.KG}(\text{fe.msk}, \text{RE}[\text{one.pk}_i])$ for $i \in [q]$. Without loss of generality, we assume $\tau_{\text{one.pk}_1} < \cdots < \tau_{\text{one.pk}_q}$, which can be ensured by relabeling if necessary.

We consider dummy keys $\text{one.pk}_0 := 0^{\ell_{\text{pk}}}$ and $\text{one.pk}_{q+1} := 1^{\ell_{\text{pk}}}$, and set $\tau_{\text{one.pk}_0} := 0^{\ell_{\text{pk}}}$ and $\tau_{\text{one.pk}_{q+1}} := 1^{\ell_{\text{pk}}}$. Let D_i be the following distribution for each $i \in \{0, 1, \dots, q\}$.

D_i :

- Sample coin $\leftarrow \{0, 1\}$ and $K \leftarrow \text{PRF.Gen}(1^\lambda)$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, \text{m}_{\text{coin}} \| K \| 1 \| \text{one.pk}_i \| 0^{\ell_{\text{ct}}})$.

We also let D_{-1} and D_{q+1} be the following distributions:

D_{-1} :

- Sample coin $\leftarrow \{0, 1\}$ and $K \leftarrow \text{PRF.Gen}(1^\lambda)$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, \text{m}_{\text{coin}} \| K \| 0 \| 0^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}})$.

D_{q+1} :

- Sample coin $\leftarrow \{0, 1\}$ and $K \leftarrow \text{PRF.Gen}(1^\lambda)$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, 0^{\ell_{\text{m}}} \| K \| 1 \| 1^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}})$.
- It holds that $\text{FE.Enc}(\text{fe.pk}, \text{m} \| K \| 0 \| 0^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}}) \approx \text{FE.Enc}(\text{fe.pk}, \text{m} \| K \| 1 \| 0^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}})$ by the adaptive security of FE since $\text{RE}[\text{one.pk}_i](\text{m} \| K \| 0 \| 0^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}}) = \text{RE}[\text{one.pk}_i](\text{m} \| K \| 1 \| 0^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}})$ for all $i \in [q]$. That is, $D_{-1} \approx D_0$. Note that we can assume that $\text{one.pk}_i \neq 0^{\ell_{\text{pk}}}$.
- It holds that $\text{FE.Enc}(\text{fe.pk}, \text{m} \| K \| 1 \| \text{one.pk}_q \| 0^{\ell_{\text{ct}}}) \approx \text{FE.Enc}(\text{fe.pk}, 0^{\ell_{\text{m}}} \| K \| 1 \| 1^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}})$ by the adaptive security of FE since $\text{RE}[\text{one.pk}_i](\text{m} \| K \| 1 \| \text{one.pk}_q \| 0^{\ell_{\text{ct}}}) = \text{RE}[\text{one.pk}_i](0^{\ell_{\text{m}}} \| K \| 1 \| 1^{\ell_{\text{pk}}} \| 0^{\ell_{\text{ct}}})$ for all $i \in [q]$. Note that we assume $\tau_{\text{one.pk}_1} < \cdots < \tau_{\text{one.pk}_q}$. That is, $D_q \approx D_{q+1}$.

We define a POVM based on D_i for $i \in \{0, \dots, q\}$.

- Let $\mathcal{P}_i = (P_i, I - P_i)$ be the following POVM.
 - Sample coin $\leftarrow \{0, 1\}$, PRF key K , and ciphertext randomness.
 - Compute $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, \text{m}_{\text{coin}} \| K \| 1 \| \text{one.pk}_i \| 0^{\ell_{\text{ct}}})$

- Run the quantum decryptor on input `fe.ct`. If the outcome is `coin`, output 1. Otherwise, output 0.

- Let $\text{Pl}_i := \text{ProjImp}(\mathcal{P}_i)$ be the projective implementation of $\mathcal{P}_i$.

We can similarly define POVM $\mathcal{P}_{-1}$ and $\mathcal{P}_{q+1}$ based on D_{-1} and D_{q+1} and projective implementation Pl_{-1} and Pl_{q+1} as above.

By [Theorem 4](#), if a quantum decryptor lies within the subspace defined by $\text{TI}_{1/2+\gamma}$, then applying Pl_{-1} to the state always yields a real-valued outcome $\beta \geq 1/2 + \gamma$. Let $b_{i,-1}$ and $b_{i,q+1}$ denote the random variables representing the outcomes of applying Pl_{-1} and Pl_{q+1} , respectively, to the i -th quantum decryptor in the post-measurement state δ^* . Then we have:

$$\Pr \left[\forall i \in [q+1], b_{i,-1} \geq \frac{1}{2} + \gamma \right] = 1, \quad (13)$$

$$\Pr \left[\forall i \in [q+1], b_{i,q+1} = \frac{1}{2} \right] = 1. \quad (14)$$

The latter equation holds since the ciphertext in Pl_{q+1} does not depend on any information about m_0 or m_1 .

Let $b_{i,0}$ and $b_{i,q}$ denote the random variables representing the outcomes of applying Pl_0 and Pl_q , respectively, to the i -th quantum decryptor in δ^* . By $D_{-1} \approx D_0$ and [Theorem 5](#), we have

$$\Pr \left[\forall i \in [q+1], b_{i,0} \geq \frac{1}{2} + \gamma - \frac{\gamma}{c} \right] = 1 - \text{negl}(\lambda), \quad (15)$$

where c is some small constant (say, $c = 10$). For the k -th quantum decryptor $\delta^*[k]$, we apply Pl_0 to obtain an outcome $b_{k,0}$. We apply projective implementation Pl_i on the leftover state to obtain an outcome $b_{k,i}$ for $i \in \{1, \dots, q\}$.

Lemma 20. *There always exists $j \in \{1, \dots, q\}$ such that $b_{1,j-1} - b_{1,j} \geq \gamma'/q$.*

Proof. First, we prove that for any inverse polynomial p' , $\Pr[b_{1,q} - b_{1,q+1} \geq p'] = \text{negl}(\lambda)$ holds. If $\Pr[b_{1,q} - b_{1,q+1} \geq p'] = \text{non-negl}(\lambda)$ for some inverse polynomial p' , by [Theorem 5](#), D_q and D_{q+1} must be distinguishable. As we saw above, $D_q \approx D_{q+1}$ by the adaptive security of FE. Hence, $b_{1,0} - b_{1,q} \geq \gamma - \frac{\gamma}{c} - \text{negl}(\lambda)$. From this and triangle inequality, we derive the claim by setting $\gamma' := \gamma - \frac{\gamma}{c} - \text{negl}(\lambda)$. $\square$

We similarly have the following lemma.

Lemma 21. *For every $i \in [q+1]$, there always exists $j \in [q]$ such that $b_{i,j-1} - b_{i,j} \geq \gamma'/q$.*

Let idx be the first index such that $b_{\text{idx},j_{\text{idx}}-1} - b_{\text{idx},j_{\text{idx}}} \geq \gamma'/q$.

Lemma 22. *There exist $\text{idx}_1 \neq \text{idx}_2$ such that $\text{idx}_1, \text{idx}_2 \in [q+1]$ and $j_{\text{idx}_1} = j_{\text{idx}_2}$ with probability 1.*

Proof. The pigeonhole principle immediately provides this lemma. $\square$

We describe a part of our reduction algorithm named $\mathcal{R}_0$ below.

Lemma 23. *$\mathcal{R}_0$ in [Figure 2](#) outputs $(\text{idx}_1, \text{idx}_2, j, b_{\text{idx}_1,j-1}, b_{\text{idx}_2,j-1})$ and $\delta^{**}[\text{idx}_1, \text{idx}_2]$ satisfies the following with probability at least $1/(2q^3)$.*

Reduction algorithm $\mathcal{R}_0$

Input: $q + 1$ quantum decryptors δ^* .

1. Sample $1 \leq \text{idx}_1 \leq \text{idx}_2 \leq q + 1$ and $j \in [q]$.
2. Apply Pl_{-1} to Pl_{j-1} to $\delta^*[\text{idx}_1]$. Let $b_{\text{idx}_1, j-1}$ be the last outcome.
3. Apply Pl_{-1} to Pl_{j-1} to $\delta^*[\text{idx}_2]$. Let $b_{\text{idx}_2, j-1}$ be the last outcome.
4. Output $(\text{idx}_1, \text{idx}_2, j, b_{\text{idx}_1, j-1}, b_{\text{idx}_2, j-1})$ and both the idx_1 -th and idx_2 -th quantum decryptors, denoted by $\delta^{**}[\text{idx}_1, \text{idx}_2]$.

Figure 2: The description of reduction $\mathcal{R}_0$

1. Applying $\text{Pl}_{j-1}^{\otimes 2}$ jointly to $\delta^*[\text{idx}_1, \text{idx}_2]$ yields $b_{\text{idx}_1, j-1}, b_{\text{idx}_2, j-1}$ with probability 1.
2. Applying $\text{Pl}_j^{\otimes 2}$ jointly to $\delta^*[\text{idx}_1, \text{idx}_2]$ yields $b_{\text{idx}_1, j}, b_{\text{idx}_2, j}$ such that

$$\Pr \left[b_{\text{idx}_1, j-1} - b_{\text{idx}_1, j} \geq \frac{\gamma'}{q} \wedge b_{\text{idx}_2, j-1} - b_{\text{idx}_2, j} \geq \frac{\gamma'}{q} \right] \geq \frac{1}{2q^3}$$

We will construct a distinguisher for D_{j-1} and D_j by using idx_1 -th and idx_2 -th decryptors $\delta^{**}[\text{idx}_1, \text{idx}_2]$.

Proof of Lemma 23. By Lemma 21, there exists a pair of indices $\text{idx}_1 < \text{idx}_2$ and an integer $j \in [q]$ such that $b_{\text{idx}_1, j-1} - b_{\text{idx}_1, j} \geq \frac{\gamma'}{q}$ and $b_{\text{idx}_2, j-1} - b_{\text{idx}_2, j} \geq \frac{\gamma'}{q}$. To exploit this, we randomly guess idx_1 and idx_2 and j after applying the sequence of projective implementations $\text{Pl}_0, \dots, \text{Pl}_q$ to each quantum decryptor. Then, we have the following bound:

$$\Pr \left[b_{\text{idx}_1, j-1} - b_{\text{idx}_1, j} \geq \frac{\gamma'}{q} \wedge b_{\text{idx}_2, j-1} - b_{\text{idx}_2, j} \geq \frac{\gamma'}{q} \right] \geq \frac{1}{\frac{q(q-1)}{2} \cdot q} \geq \frac{1}{q^3}. \quad (16)$$

Since the repeated projective implementations on disjoint quantum decryptors, they commute. Therefore, we achieve the same probability bound even if we restrict the measurements to only the idx_1 -th and idx_2 -th decryptors, omitting the other $(q-1)$ ones. Hence, we obtain:

$$\Pr_{\text{RM}(\delta^*)} [b_{\text{idx}_1, j-1} - b_{\text{idx}_1, j} \geq \frac{\gamma'}{q} \wedge b_{\text{idx}_2, j-1} - b_{\text{idx}_2, j} \geq \frac{\gamma'}{q}] \geq \frac{1}{q^3}. \quad (17)$$

The difference between the two inequalities lies in how the values $b_{\text{idx}_1, j-1}, b_{\text{idx}_1, j}, b_{\text{idx}_2, j-1}, b_{\text{idx}_2, j}$ are sampled. Algorithm RM first runs $\mathcal{R}_0$ to get $(\text{idx}_1, \text{idx}_2, j, b_{\text{idx}_1, j-1}, b_{\text{idx}_2, j-1})$ and $\delta^{**}[\text{idx}_1, \text{idx}_2]$, then applies Pl_j on idx_1 -th and idx_2 -th registers.

If Lemma 23 does not hold, $(\text{idx}_1, \text{idx}_2, j, b_{\text{idx}_1, j-1}, b_{\text{idx}_2, j-1})$ and $\delta^{**}[\text{idx}_1, \text{idx}_2]$ satisfy condition (2) in Lemma 23 with probability less than $1/2q^3$. Hence,

$$\Pr_{\text{RM}(\delta^*)} [b_{\text{idx}_1, j-1} - b_{\text{idx}_1, j} \geq \frac{\gamma'}{q} \wedge b_{\text{idx}_2, j-1} - b_{\text{idx}_2, j} \geq \frac{\gamma'}{q}]$$

is strictly smaller than $1/2q^3 + 1/2q^3$, which is a contradiction. (The test projector commutes with rest of RM.) $\square$

Algorithm RM

Input: $q + 1$ quantum decryptors δ^* .

1. Sample $1 \leq \text{idx}_1 \leq \text{idx}_2 \leq q + 1$ and $j \in [q]$.
2. Apply repeated projective measurement Pl_{-1} to Pl_{j-1} to $\delta^*[\text{idx}_1]$. Let $b_{\text{idx}_1, j-1}$ and $b_{\text{idx}_1, j}$ be the last two outcomes.
3. Apply repeated projective measurement Pl_{-1} to Pl_{j-1} to $\delta^*[\text{idx}_2]$. Let $b_{\text{idx}_2, j-1}$ and $b_{\text{idx}_2, j}$ be the last two outcomes.
4. Output $(\text{idx}_1, \text{idx}_2, j, b_{\text{idx}_1, j-1}, b_{\text{idx}_2, j-1})$.

Figure 3: The description of RM

Lemma 24. *Let $q = \text{poly}(\lambda)$ be any polynomial. Let $\text{pk} := \text{fe.pk}$, where $(\text{fe.pk}, \text{fe.msk}) \leftarrow \text{FE.Setup}(1^\lambda)$. Let $\text{sk}_i = (\text{one.sk}_i, \text{fe.fsk}_i)$ be the i -th decryption key generated by $\text{KG}(\text{sk})$, where $\text{sk} = \text{fe.msk}$. For any $j \in [q]$, let Pl_{j-1} and Pl_j be defined at the beginning of the proof. If there exist inverse polynomials $\alpha_1(\cdot), \alpha_2(\cdot)$ and an quantum algorithm $\mathcal{B}$ that takes $(\text{sk}_1, \dots, \text{sk}_q, \text{pk})$ outputs δ such that with probability at least α_1 , δ satisfies the following:*

1. *There exist $b_{j-1} \in (0, 1]$, applying Pl_{j-1} on δ always outputs b_{j-1} .*
2. *Let the outcome of applying Pl_j on δ be b_j . We have $\Pr[b_{j-1} - b_j > \gamma'/q] > \alpha_2$.*

Then, there exist another inverse polynomial $\beta(\cdot)$ and an efficient quantum algorithm $\mathcal{A}$ that takes δ and distinguishes D_j from D_{j-1} with probability greater than β .

Proof of Lemma 24. Recall that $\text{Pl}_j := \text{ProjImp}(\mathcal{P}_j)$ and $\mathcal{P}_j$ is a POVM based on D_j . Hence, we immediately obtain the lemma from [Theorem 5](#). $\square$

If we set $\alpha_1 = \alpha_2 := 1/2q^3$, and let $\mathcal{B}$ be the $\mathcal{R}_0$ in [Figure 2](#) and $\rho := \delta^{**}[\text{idx}_1]$, we have an algorithm that takes $\delta^{**}[\text{idx}_1]$ and distinguishes D_j from D_{j-1} .

We define the following sub-hybrid distributions:

D_{i-1}^1 :

- Sample $\text{coin} \leftarrow \{0, 1\}$, $K \leftarrow \text{PRF.Gen}(1^\lambda)$, and $\text{one.ct}^* \leftarrow \text{One.Enc}(\text{one.pk}_i, m_{\text{coin}}; F_K(\text{one.pk}_i))$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, m_{\text{coin}} \| K \| 2 \| \text{one.pk}_i \| \text{one.ct}^*)$.

D_{i-1}^2 :

- Sample $\text{coin} \leftarrow \{0, 1\}$, $K \leftarrow \text{PRF.Gen}(1^\lambda)$, and $\text{one.ct}^* \leftarrow \text{One.Enc}(\text{one.pk}_i, m_{\text{coin}}; F_K(\text{one.pk}_i))$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, m_{\text{coin}} \| K_{\neq \text{one.pk}_i} \| 2 \| \text{one.pk}_i \| \text{one.ct}^*)$, where $K_{\neq \text{one.pk}_i} \leftarrow \text{Puncture}(K, \{\text{one.pk}_i\})$.

D_{i-1}^3 :

- Sample $\text{coin} \leftarrow \{0, 1\}$, $r \leftarrow \mathcal{R}_{\text{One}}$, and $\text{one.ct}^* \leftarrow \text{One.Enc}(\text{one.pk}_i, m_{\text{coin}}; r)$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, m_{\text{coin}} \| K_{\neq \text{one.pk}_i} \| 2 \| \text{one.pk}_i \| \text{one.ct}^*)$, where $K_{\neq \text{one.pk}_i} \leftarrow \text{Puncture}(K, \{\text{one.pk}_i\})$.

D_i^{-1} :

- Sample $K \leftarrow \text{PRF.Gen}(1^\lambda)$ and $\text{one.ct}^* \leftarrow \text{One.Enc}(\text{one.pk}_i, 0^{\ell_m}; F_K(\text{one.pk}_i))$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, m_{\text{coin}} \| K \| 2 \| \text{one.pk}_i \| \text{one.ct}^*)$.

D_i^{-2} :

- Sample $K \leftarrow \text{PRF.Gen}(1^\lambda)$ and $\text{one.ct}^* \leftarrow \text{One.Enc}(\text{one.pk}_i, 0^{\ell_m}; F_K(\text{one.pk}_i))$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, m_{\text{coin}} \| K_{\neq \text{one.pk}_i} \| 2 \| \text{one.pk}_i \| \text{one.ct}^*)$, where $K_{\neq \text{one.pk}_i} \leftarrow \text{Puncture}(K, \{\text{one.pk}_i\})$.

D_i^{-3} :

- Sample $r \leftarrow \mathcal{R}_{\text{One}}$ and $\text{one.ct}^* \leftarrow \text{One.Enc}(\text{one.pk}_i, 0^{\ell_m}; r)$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, m_{\text{coin}} \| K_{\neq \text{one.pk}_i} \| 2 \| \text{one.pk}_i \| \text{one.ct}^*)$, where $K_{\neq \text{one.pk}_i} \leftarrow \text{Puncture}(K, \{\text{one.pk}_i\})$.

Lemma 25. *If there exist a distinguisher for D_{j-1} and D_j , there also exist a distinguisher for D_{j-1}^3 and D_j^{-3} .*

Proof of Lemma 25. By FE security we have $D_{i-1} \approx D_{i-1}^1$. By FE security and punctured correctness, we have $D_{i-1}^1 \approx D_{i-1}^2$. By puncturable PRF security, we have $D_{i-1}^2 \approx D_{i-1}^3$.

By FE security we have $D_i \approx D_i^{-1}$. By FE security and punctured correctness, we have $D_i^{-1} \approx D_i^{-2}$. By puncturable PRF security, we have $D_i^{-2} \approx D_i^{-3}$.

Hence, we can construct a distinguisher for D_{j-1}^3 and D_j^{-3} from a distinguisher for D_{j-1} and D_j . $\square$

The rest of our reduction algorithm $\mathcal{R}_x$ takes $\delta^{**}[\text{id}x_x]$ for $x \in \{1, 2\}$ and constructs a distinguisher for D_{j-1}^3 and D_j^{-3} . From Lemmas 23 to 25, $\mathcal{R}_1$ constructs a successful distinguisher $\mathbf{qD}_{\text{id}x_1}$ for D_{j-1}^3 and D_j^{-3} from $\delta^{**}[\text{id}x_1]$.

We show that after our reduction algorithm obtains $\mathcal{D}_{\text{id}x_1}$, the other decryptor becomes $\delta'[\text{id}x_2]$, which still satisfies the conditions in Lemma 24. To derive a contradiction, suppose instead that $\delta'[\text{id}x_2]$ does not satisfy the conditions in Lemma 24.

Note that applying Pl_{j-1} to $\delta'[\text{id}x_2]$ always yields $b_{\text{id}x_2, j-1}$ since constructing the distinguisher from $\delta^{**}[\text{id}x_1]$ register does not affect the support of $\delta'[\text{id}x_2]$. Hence, condition (2) in Lemma 24 cannot be satisfied under this assumption. We define two events:

SD₁: We have a successful distinguisher from $\delta^{**}[\text{id}x_1]$.

G₂: We apply Pl_j on $\delta^{**}[\text{id}x_2]$ to get $b_{\text{id}x_2, j}$ and it holds that $b_{\text{id}x_2, j} < b_{\text{id}x_2, j-1} - \gamma'/q$.

Our assumption implies that $\Pr[\mathbf{SD}_1 \wedge \mathbf{G}_2]$ is negligible. However, this leads to a contradiction.

By the condition (2) of Lemma 23, we have that $\Pr[\mathbf{G}_2]$ is non-negligible. Conditioned on $\mathbf{G}_2$, $\text{id}x_1$ -th decryptor becomes $\delta'[\text{id}x_1]$. Here, $\delta'[\text{id}x_1]$ must satisfy both conditions in Lemma 24. Otherwise, condition (2) in Lemma 23 cannot be satisfied. That is, $\Pr[\mathbf{G}_1 \mid \mathbf{G}_2]$ is non-negligible. This implies $\Pr[\mathbf{SD}_1 \mid \mathbf{G}_2]$ is non-negligible.

Hence, $\Pr[\mathbf{G}_2 \mid \mathbf{SD}_1]$ must be non-negligible since $\Pr[\mathbf{G}_2 \mid \mathbf{SD}_1] = \frac{\Pr[\mathbf{G}_2 \wedge \mathbf{SD}_1]}{\Pr[\mathbf{SD}_1]} = \frac{\Pr[\mathbf{SD}_1 \mid \mathbf{G}_2] \Pr[\mathbf{G}_2]}{\Pr[\mathbf{SD}_1]}$. This contradicts with the assumption that $\Pr[\mathbf{SD}_1 \wedge \mathbf{G}_2]$ is negligible. Thus, $\Pr[\mathbf{SD}_1 \wedge \mathbf{G}_2]$ is non-negligible. This implies $\Pr[\mathbf{SD}_1 \wedge \mathbf{SD}_2]$ is non-negligible.

Hence, $\mathcal{R}_2$ can obtain a successful distinguisher $\mathbf{qD}_{\text{id}x_2}$ for D_{j-1}^3 and D_j^{-3} from $\delta'[\text{id}x_2]$ as well.

Now, we define $D_j^{\pm 3}$ as:

$D_j^{\pm 3}$:

- Let $m_0^* := m_{\text{coin}}$ and $m_1^* := 0^{\ell_m}$.
- Sample $b \leftarrow \{0, 1\}$, $r \leftarrow \mathcal{R}_{\text{One}}$, and $\text{one.ct}^* \leftarrow \text{One.Enc}(\text{one.pk}_i, m_b^*; r)$.
- Let the distribution be $\text{fe.ct} \leftarrow \text{FE.Enc}(\text{fe.pk}, m_b^* \| K_{\neq \text{one.pk}_j} \| 2 \| \text{one.pk}_j \| \text{one.ct}^*)$, where $K_{\neq \text{one.pk}_j} \leftarrow \text{Puncture}(K, \{\text{one.pk}_j\})$.

In the case of $b = 1$, $D_j^{\pm 3}$ is the same as D_j^{-3} . In the case of $b = 0$, $D_j^{\pm 3}$ is the same as D_{j-1}^3 where coin is fixed in advance. By averaging argument, if there exists a distinguisher $q\mathcal{D}_{\text{id}_{x_x}}$ for D_j^{-3} and D_{j-1}^3 with an inverse polynomial advantage, there also exists a distinguisher $q\mathcal{D}'_{\text{id}_{x_x}}$ for the case $b = 0, 1$ of $D_j^{\pm 3}$ with an inverse polynomial advantage. Let $1/2 + \gamma^*$ be the distinguishing advantage of $q\mathcal{D}'_{\text{id}_{x_x}}$ for the case $b = 0, 1$ of $D_j^{\pm 3}$.

Hence, if we apply $\Pi_{1/2+\gamma^*}(\mathcal{P}_{D_j^{\pm 3}})$ to $q\mathcal{D}_{\text{id}_{x_x}}$, then we obtain a γ^* -good decryptor with respect to one.pk_j and $(m_{\text{coin}}, 0^{\ell_m})$. The reduction runs $\mathcal{R}_0$ on δ^* , then $\mathcal{R}_x$ constructs a distinguisher $q\mathcal{D}'_{\text{id}_{x_x}}$ from $\delta^{**}[\text{id}_{x_x}]$, applies $\Pi_{1/2+\gamma^*}(\mathcal{P}_{D_j^{\pm 3}})$ to $q\mathcal{D}'_{\text{id}_{x_x}}$. Let the resulting γ^* -good decryptor be $q\mathcal{D}_x^*$. The reduction outputs a pair of messages $(m_{\text{coin}}, 0^{\ell_m})$ and two quantum decryptors $(q\mathcal{D}_1^*, q\mathcal{D}_2^*)$.

Hence, we can construct an adversary $\mathcal{B}$ for One from the adversary $\mathcal{A}$ for SDE. $\mathcal{B}$ is given a public key $\text{one.pk}^\dagger$ and a quantum decryption key $\text{one.sk}^\dagger$, generates $(\text{fe.pk}, \text{fe.msk}) \leftarrow \text{FE.Setup}(1^\lambda)$ and $(\text{one.pk}_i, \text{one.sk}_i) \leftarrow \text{One.Setup}(1^\lambda)$ for $i \in [q] \setminus \{j\}$. Note that we sample random tags such that $\tau_{\text{one.pk}_1} < \dots < \tau_{\text{one.pk}_q}$ by appropriately renaming tags. $\mathcal{B}$ can run $\mathcal{A}$ since $\mathcal{B}$ has $(\text{fe.pk}, \text{fe.msk})$, $\{(\text{one.pk}_i, \text{one.sk}_i)\}_{i \in [q] \setminus \{j\}}$, $\text{one.pk}_j := \text{one.pk}^\dagger$, and $\text{one.sk}_j := \text{one.sk}^\dagger$. The rest of $\mathcal{B}$ is the same as the reduction above except that sample $j \in [q]$ since $\mathcal{B}$ had already sampled it. Hence, $\mathcal{B}$ can output a pair of messages $(m_{\text{coin}}, 0^{\ell_m})$ and two quantum decryptors $(q\mathcal{D}_1^*, q\mathcal{D}_2^*)$ such that $q\mathcal{D}_x^*$ is γ^* -good decryptor with respect to $\text{one.pk}_j = \text{one.pk}^\dagger$ and $(m_{\text{coin}}, 0^{\ell_m})$. This completes the proof of strong anti-piracy security.

Remark 2. We can prove the strong search anti-piracy security similarly to the proof of the strong anti-piracy security given above. The main differences are [Equations \(13\) and \(14\)](#). Instead of them, we have

$$\Pr[\forall i \in [q+1], b_{i,-1} \geq 1/2^{\ell_m} + \gamma] = 1, \quad (18)$$

$$\Pr[\forall i \in [q+1], b_{i,q+1} = 1/2^{\ell_m}] = 1. \quad (19)$$

Hence, there always exists $j \in \{1, \dots, q\}$ such that $b_{1,j-1} - b_{1,j} \geq \tilde{\gamma}/q$ as [Lemma 20](#). The rest of the proof is mostly the same as before.

8.4 Collusion-Resistant Identical-Challenge Search Security

It is straightforward to show that collusion-resistant strong search anti-piracy security implies plain collusion-resistant search anti-piracy security in the independent-challenge setting, where the challenge ciphertexts given to each party are generated independently, following a similar claim in the single-key setting [\[CLLZ21\]](#). On the other hand, it is not immediately clear whether this implication also holds in the identical-challenge setting, where all parties receive the same challenge ciphertext. In this section, we show that this implication indeed holds, relying on a technique similar to the one used in [\[AKL23\]](#).

We first give the definition of collusion-resistant identical-challenge search security.

Definition 24 (Collusion-Resistant Identical-Challenge Search Anti-Piracy). Let $\text{SDE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$ be a public-key SDE scheme. We consider the collusion-resistant identical-challenge search-type anti-piracy game $\text{Exp}_{\text{SDE}, \mathcal{A}}^{\text{idn-chal-search}}(\lambda, \gamma)$ between the challenger and an adversary $\mathcal{A}$ below.

1. The challenger generates $(\text{pk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda)$ and sends $(1^\lambda, \text{pk})$ to $\mathcal{A}$.
2. $\mathcal{A}$ sends 1^q to the challenger, then the challenger generates $\text{sk}_i \leftarrow \text{KG}(\text{msk})$ for $i \in [q]$, and sends $(\text{sk}_1, \dots, \text{sk}_q)$ to $\mathcal{A}$.
3. $\mathcal{A}$ outputs $(q+1)$ (possibly entangled) quantum decryptors $\mathbf{qD}_1 = (\delta[R_1], \mathbf{U}_1), \dots, \mathbf{qD}_{q+1} = (\delta[R_{q+1}], \mathbf{U}_{q+1})$, where δ is a quantum state over registers $R_1, \dots, R_{q+1}$, and each $\mathbf{U}_i$ is a unitary quantum circuit.
4. The challenger samples $\mathbf{m} \leftarrow \mathcal{M}$ and generates $\text{ct} \leftarrow \text{Enc}(\text{pk}, \mathbf{m})$. The challenger runs quantum decryptors $\mathbf{qD}_i$ on ct for all $i \in [q+1]$, and checks that all $\mathbf{qD}_i$ outputs $\mathbf{m}$. If so, the challenger outputs 1; otherwise outputs 0.

We say that SDE is collusion-resistant identical-challenge search γ -anti-piracy secure if for any QPT adversary $\mathcal{A}$, it satisfies that

$$\text{Adv}_{\text{SDE}, \mathcal{A}}^{\text{idn-chal-search}}(\lambda, \gamma) := \Pr \left[\text{Exp}_{\text{SDE}, \mathcal{A}}^{\text{idn-chal-search}}(\lambda, \gamma) = 1 \right] \leq \frac{1}{|\mathcal{M}|} + \gamma(\lambda) + \text{negl}(\lambda). \quad (20)$$

We say that SDE is identical-challenge search anti-piracy secure if it is identical-challenge search γ -anti-piracy secure for any inverse polynomial γ .

Now we state and prove our theorem.

Theorem 16. If SDE is strong-search anti-piracy secure and $1/|\mathcal{M}|$ is negligible, it is also identical-challenge search anti-piracy secure.

Proof. Let $\mathcal{R}_{\text{Enc}}$ be encryption randomness space of SDE. We set $\mathcal{R} := \mathcal{R}_{\text{Enc}} \times \mathcal{M}$. Let $\mathbf{D}$ be a uniform distribution over $\mathcal{R}$. Let $\tilde{\mathbf{D}}$ be a distribution that samples a uniformly random r from $\mathcal{R}$ and outputs $(r, \dots, r) \in \mathcal{R}^{q+1}$. Let δ be a mixed $(q+1)$ -partite state. We consider the following POVM element $M_r^{(i)}$ acting on the i -th register of δ :

- Parse $r = (r_{\text{Enc}}, \mathbf{m})$.
- Compute $\text{ct} := \text{Enc}(\text{pk}, \mathbf{m}; r_{\text{Enc}})$.
- Run the quantum decryptor on input ct . If the outcome is $\mathbf{m}$, output 1.

We define the following quantity:

$$p_{\text{idn}} := \mathbb{E}_{(r, \dots, r) \leftarrow \tilde{\mathbf{D}}} \text{Tr} \left[(M_r^{(1)} \otimes \dots \otimes M_r^{(q+1)}) \delta \right],$$

where for each $r \in \mathcal{R}$, we have $\mathbf{0} \leq M_r^{(i)} \leq \mathbf{I}$ for all $i \in [q+1]$. Here, δ is considered to be the quantum decryptors output by $\mathcal{A}$. We will show

$$p_{\text{idn}} \leq (q+2)\gamma(\lambda) + \text{negl}(\lambda) \quad (21)$$

if SDE is strong search γ -anti-piracy secure and $\frac{1}{|\mathcal{M}|}$ is negligible.

By the Naimark Dilation theorem, we assume without loss of generality that all measurements $M_r^{(i)}$ are projective for each $i \in [q+1]$. We can also assume that $\delta = |\psi\rangle\langle\psi|$ is pure. The general (mixed-state) case follows by convexity.

To prove Equation (21), we define

$$P^{(i)} := \mathbb{E}_{r \leftarrow \mathbf{D}} M_r^{(i)},$$

and let $|\psi\rangle = \sum_{i_1, \dots, i_{q+1}} \alpha_{i_1, \dots, i_{q+1}} |\phi_{i_1}^{(1)}\rangle \cdots |\phi_{i_{q+1}}^{(q+1)}\rangle$ be the spectral decomposition of $|\psi\rangle$ with respect to $P^{(1)} \otimes \cdots \otimes P^{(q+1)}$, where for each $j \in [q+1]$, $|\phi_{i_j}^{(j)}\rangle$ is an eigenvector of $P^{(j)}$ corresponding to eigenvalue $\mu_{i_j}^{(j)}$. To analyze p_{idn} , we set $\eta := \gamma(\lambda) + \text{negl}(\lambda)$ and define

$$\begin{aligned} |\psi^{(1)}\rangle &= \sum_{\substack{i_1: \mu_{i_1}^{(1)} \leq \eta \\ i_2: \mu_{i_2}^{(2)} > \eta \\ \vdots \\ i_{q+1}: \mu_{i_{q+1}}^{(q+1)} > \eta}} \alpha_{i_1, \dots, i_{q+1}} |\phi_{i_1}^{(1)}\rangle \cdots |\phi_{i_{q+1}}^{(q+1)}\rangle, \\ &\vdots \\ |\psi^{(j)}\rangle &= \sum_{\substack{i_1, \dots, i_{j-1} \\ i_j: \mu_{i_j}^{(j)} \leq \eta \\ i_{j+1}: \mu_{i_{j+1}}^{(j+1)} > \eta \\ \vdots \\ i_{q+1}: \mu_{i_{q+1}}^{(q+1)} > \eta}} \alpha_{i_1, \dots, i_{q+1}} |\phi_{i_1}^{(1)}\rangle \cdots |\phi_{i_{q+1}}^{(q+1)}\rangle, \\ &\vdots \\ |\psi^{(q+1)}\rangle &= \sum_{\substack{i_1, \dots, i_q \\ i_{q+1}: \mu_{i_{q+1}}^{(q+1)} \leq \eta}} \alpha_{i_1, \dots, i_{q+1}} |\phi_{i_1}^{(1)}\rangle \cdots |\phi_{i_{q+1}}^{(q+1)}\rangle, \\ |\psi_S\rangle &= \sum_{\substack{i_1: \mu_{i_1}^{(1)} > \eta \\ \vdots \\ i_{q+1}: \mu_{i_{q+1}}^{(q+1)} > \eta}} \alpha_{i_1, \dots, i_{q+1}} |\phi_{i_1}^{(1)}\rangle \cdots |\phi_{i_{q+1}}^{(q+1)}\rangle, \end{aligned}$$

so that $|\psi\rangle = \sum_{j=1}^{q+1} |\psi^{(j)}\rangle + |\psi_S\rangle$. By the definition of $|\psi_S\rangle$ and the strong-search anti-piracy security of SDE, we have

$$\|\psi_S\|^2 \leq \text{negl}(\lambda). \quad (22)$$

For notational convenience, let

$$M_r^* := M_r^{(1)} \otimes \cdots \otimes M_r^{(q+1)}, \quad (23)$$

$$M_r^{(-i)} := \overbrace{\mathbf{I} \otimes \cdots \otimes \mathbf{I}}^{i-1} \otimes M_r^{(i)} \otimes \overbrace{\mathbf{I} \cdots \otimes \mathbf{I}}^{q+1-i}, \quad (24)$$

$$P^{(-i)} := \overbrace{\mathbf{I} \otimes \cdots \otimes \mathbf{I}}^{i-1} \otimes P^{(i)} \otimes \overbrace{\mathbf{I} \cdots \otimes \mathbf{I}}^{q+1-i}. \quad (25)$$

Then, we have

$$p_{\text{iden}} = \mathbb{E}_{(r, \dots, r) \leftarrow \tilde{\mathcal{D}}} \left\| (M_r^*) \left(|\psi^{(1)}\rangle + \dots + |\psi^{(q+1)}\rangle + |\psi_S\rangle \right) \right\|^2$$

$$\leq \mathbb{E}_{(r, \dots, r) \leftarrow \tilde{\mathcal{D}}} (q+2) \left(\left\| (M_r^*) |\psi^{(1)}\rangle \right\|^2 + \dots + \left\| (M_r^*) |\psi^{(q+1)}\rangle \right\|^2 + \left\| (M_r^*) |\psi_S\rangle \right\|^2 \right) \quad (26)$$

$$\leq \mathbb{E}_{(r, \dots, r) \leftarrow \tilde{\mathcal{D}}} (q+2) \left(\left\| (M_r^{(-1)}) |\psi^{(1)}\rangle \right\|^2 + \dots + \left\| (M_r^{(-(q+1))}) |\psi^{(q+1)}\rangle \right\|^2 + \left\| |\psi_S\rangle \right\|^2 \right)$$

$$= \mathbb{E}_{r, \dots, r \leftarrow \mathcal{D}} (q+2) \left(\left\| (M_r^{(-1)}) |\psi^{(1)}\rangle \right\|^2 + \dots + \left\| (M_r^{(-(q+1))}) |\psi^{(q+1)}\rangle \right\|^2 + \left\| |\psi_S\rangle \right\|^2 \right)$$

$$= (q+2) \left(\langle \psi^{(1)} | P^{(-1)} | \psi^{(1)} \rangle + \dots + \langle \psi^{(q+1)} | P^{(-(q+1))} | \psi^{(q+1)} \rangle + \left\| |\psi_S\rangle \right\|^2 \right)$$

$$= (q+2) \left(\eta \left(\left\| \psi^{(1)} \right\|^2 + \dots + \left\| \psi^{(q+1)} \right\|^2 \right) + \left\| |\psi_S\rangle \right\|^2 \right)$$

$$\leq (q+2) \left(\eta \left(\left\| \psi^{(1)} \right\|^2 + \dots + \left\| \psi^{(q+1)} \right\|^2 \right) + \text{negl}(\lambda) \right) \quad (27)$$

$$\leq (q+2)(\gamma(\lambda) + \text{negl}(\lambda))$$

$$= (q+2)\gamma(\lambda) + \text{negl}(\lambda).$$

Equation (26) is derived by Cauchy-Schwartz inequality. Equation (27) is derived by Equation (22). By the definition,

$$\text{Adv}_{\text{SDE}, \mathcal{A}}^{\text{iden-chal-search}}(\lambda, \gamma) = p_{\text{iden}} \leq (q+2)\gamma(\lambda) + \text{negl}(\lambda).$$

Hence, if SDE is strong-search γ -anti-piracy secure, then SDE is identical-challenge search-type γ' -anti-piracy secure, where $\gamma' = (q+2)\gamma$. This completes the proof. $\square$

8.5 Multi-Copy Secure Construction

Here, we apply our compiler to convert a collusion-resistant SDE scheme into multi-copy secure one.

First, we define multi-copy security of SDE.

Definition 25 (Multi-Copy Anti-Piracy). *Let $\text{SDE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$ be a public-key SDE scheme where KG has classically determined outputs (Definition 6). We define its multi-copy strong anti-piracy security, multi-copy strong search anti-piracy security, and multi-copy identical-challenge search anti-piracy security analogously to their collusion-resistant counterparts (Definitions 21, 22 and 24), except that the adversary is given q exact copies of the same secret key generated under the same randomness, rather than independently generated keys.*

We observe that, in the construction given in Section 8.2, the key generation algorithm has classically determined outputs (Definition 6) if the underlying single-key scheme does. Note that there exists a single-key SDE scheme that satisfies the property by Theorem 14. Therefore, we can apply Theorem 7 to compile it into a multi-copy secure scheme. We omit the description of the scheme, as it is very similar to that of the quantum coin scheme in Section 6.1. As a result, we obtain the following theorem.

Theorem 17. *Assuming the existence of polynomially secure indistinguishability obfuscation and one-way functions, there exists a public-key SDE scheme that satisfies*

- multi-copy strong anti-piracy security
- multi-copy strong search anti-piracy security
- multi-copy identical-challenge search anti-piracy security.

9 Unclonable Encryption

In this section, construct an unclonable encryption (UE) scheme that satisfies multi-copy search security.

By [Theorem 7](#), the construction of a multi-copy secure UE scheme is reduced to constructing a multi-challenge secure variant, where the adversary is given multiple ciphertexts of the same challenge message under independent encryption randomness.¹⁰ Thus, we start by constructing multi-challenge secure scheme and then convert it into multi-copy secure one.

9.1 Definitions

Definition 26 (UE (Syntax)). *A UE scheme with the message space $\mathcal{M}$ is a tuple of QPT algorithms $(\text{KG}, \text{Enc}, \text{Dec})$.*

$\text{KG}(1^\lambda) \rightarrow (\text{ek}, \text{dk})$: *The key generation algorithm takes as input the security parameter 1^λ and outputs an encryption key ek and a decryption key dk .*

$\text{Enc}(\text{ek}, \text{m}) \rightarrow \text{ct}$: *The encryption algorithm takes as input ek and a plaintext $\text{m} \in \mathcal{M}$ and outputs a ciphertext ct .*

$\text{Dec}(\text{dk}, \text{ct}) \rightarrow \text{m}'$: *The decryption algorithm takes as input dk and ct and outputs a plaintext $\text{m}' \in \mathcal{M}$ or $\perp$.*

Decryption correctness: *For any $\text{m} \in \mathcal{M}$, it holds that*

$$\Pr \left[\text{Dec}(\text{dk}, \text{ct}) = \text{m} \mid \begin{array}{l} (\text{ek}, \text{dk}) \leftarrow \text{KG}(1^\lambda) \\ \text{ct} \leftarrow \text{Enc}(\text{ek}, \text{m}) \end{array} \right] = 1 - \text{negl}(\lambda). \quad (28)$$

Remark 3. *The UE syntax above is slightly different from the standard one since it uses two different secret keys ek and dk . We believe this relaxation could be acceptable. In addition, when we compile a multi-challenge UE scheme into multi-copy one, this limitation can be removed by a simple one-time padding technique (see [Appendix A](#)).*

Definition 27 (Multi-Challenge Search Security). *Let $\text{UE} = \text{UE}(\text{KG}, \text{Enc}, \text{Dec})$ be a UE scheme. We consider the following security experiment $\text{Exp}_{\text{UE}, \mathcal{A}}^{\text{multi-chal-search}}(\lambda)$, where $\mathcal{A} = (\mathcal{A}_0, \mathcal{A}_1, \dots, \mathcal{A}_{q+1})$.*

1. *The challenger generates $(\text{ek}, \text{dk}) \leftarrow \text{UE.KG}(1^\lambda)$ and sends 1^λ to $\mathcal{A}_0$.*
2. *$\mathcal{A}_0$ sends 1^q to the challenger.*
3. *The challenger samples $\text{m} \leftarrow \mathcal{M}$, generates $\text{ct}_i \leftarrow \text{UE.Enc}(\text{ek}, \text{m})$ for all $i \in [q]$, and sends $\{\text{ct}_i\}_{i \in [q]}$ to $\mathcal{A}_0$.*

¹⁰Multi-challenge security is conceptually equivalent to collusion resistance, but we retain the term multi-challenge as it is more natural in the context of UE.

4. $\mathcal{A}_0$ creates a $(q+1)$ -partite state δ over registers $R_1, \dots, R_{q+1}$, and sends R_i to $\mathcal{A}_i$ for each $i \in [q+1]$.
5. The challenger sends dk to $\mathcal{A}_i$ for each $i \in [q+1]$. Each $\mathcal{A}_i$ outputs $\mathbf{m}'_i$. If $\mathbf{m}'_i = \mathbf{m}$ for all $i \in [q+1]$, the challenger outputs 1, otherwise outputs 0.

We say that UE is multi-challenge search secure UE scheme if for any QPT $\mathcal{A}$, it holds that

$$\text{Adv}_{\text{UE}, \mathcal{A}}^{\text{multi-chal-search}}(\lambda) := \Pr \left[\text{Exp}_{\text{UE}, \mathcal{A}}^{\text{multi-chal-search}}(\lambda) = 1 \right] \leq \frac{1}{|\mathcal{M}|} + \text{negl}(\lambda). \quad (29)$$

The building block is an SDE scheme $\text{SDE} = \text{SDE}(\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$. We construct a UE scheme $\text{UE} = \text{UE}(\text{KG}, \text{Enc}, \text{Dec})$ as follows.

Definition 28 (Multi-Copy Search Security). *Let $\text{UE} = \text{UE}(\text{KG}, \text{Enc}, \text{Dec})$ be a UE scheme such that $\text{UE}.\text{Enc}$ has classically determined outputs (Definition 6). We define multi-copy search security of UE analogously to its multi-challenge counterpart (Definition 27) except that the adversary is given q exact copies of the same ciphertext generated under the same randomness, rather than independently generated ciphertexts.*

9.2 From Collusion-Resistant SDE to Multi-Challenge UE

Here, we present a conversion from SDE to UE. Assuming that the SDE satisfies collusion-resistant identical-challenge search anti-piracy security, the following straightforward conversion, where we simply switch the roles of ciphertexts and keys, works. Let $\text{SDE} = \text{SDE}(\text{Setup}, \text{KG}, \text{Enc}, \text{Dec})$ be an SDE scheme with message space $\{0, 1\}^\ell$. Then we construct an UE scheme $\text{UE} = \text{UE}(\text{KG}, \text{Enc}, \text{Dec})$ with message space $\{0, 1\}^\ell$ as follows.

$\text{UE.KG}(1^\lambda)$:

- Generate $(\text{sde.pk}, \text{sde.msk}) \leftarrow \text{SDE.Setup}(1^\lambda)$.
- Choose $s \leftarrow \{0, 1\}^\ell$.
- Generate $\text{sde.ct} \leftarrow \text{SDE.Enc}(\text{sde.pk}, s)$.
- Output $\text{ue.ek} := (\text{sde.msk}, s)$ and $\text{ue.dk} := \text{sde.ct}$.

$\text{UE.Enc}(\text{ue.ek}, m)$:

- Parse $\text{ue.ek} = (\text{sde.msk}, s)$.
- Generate $\text{sde.sk} \leftarrow \text{SDE.KG}(\text{sde.msk})$.
- Output $\text{ue.ct} := (\text{sde.sk}, m \oplus s)$

$\text{UE.Dec}(\text{ue.dk}, \text{ue.ct})$:

- Parse $\text{ue.dk} = \text{sde.ct}$ and $\text{ue.ct} = (\text{sde.sk}, \mu)$.
- Compute $s' \leftarrow \text{SDE.Dec}(\text{sde.sk}, \text{sde.ct})$.
- Output $\mathbf{m}' := \mu \oplus s'$.

We now state our theorem and prove it.

Theorem 18. *If SDE is collusion-resistant identical-challenge search anti-piracy secure, then UE is multi-challenge search secure UE.*

Proof. We construct an adversary $\mathcal{A}_{\text{SDE}}$ for SDE by using an adversary $\mathcal{A}_{\text{UE}} = (\mathcal{A}_{\text{UE},0}, \mathcal{A}_{\text{UE},1}, \dots, \mathcal{A}_{\text{UE},q+1})$ for UE. $\mathcal{A}_{\text{SDE}}$ proceeds as follows.

1. $\mathcal{A}_{\text{SDE}}$ is given $(1^\lambda, \text{sde.pk})$, forwards 1^λ to $\mathcal{A}_{\text{UE},0}$, and receives 1^q .
2. $\mathcal{A}_{\text{SDE}}$ sends 1^q to its challenger, receives $(\text{sde.sk}_1, \dots, \text{sde.sk}_q)$.
3. $\mathcal{A}_{\text{SDE}}$ samples a uniformly random message $t \leftarrow \{0, 1\}^\ell$, set $\text{ue.ct}_i := (\text{sde.sk}_i, t)$ for all $i \in [q]$, and sends $(\text{ue.ct}_1, \dots, \text{ue.ct}_q)$ to $\mathcal{A}_{\text{UE},0}$.
4. When $\mathcal{A}_{\text{UE},0}$ creates $(q+1)$ -partite state δ over registers $R_1, \dots, R_{q+1}$, $\mathcal{A}_{\text{SDE}}$ constructs the following quantum decryptors $\mathbf{qD}_i$:
 - Receive an input ue.dk , interpret it as an SDE ciphertext $\text{sde.ct} := \text{ue.dk}$, and run $\mathbf{m}'_i \leftarrow \mathcal{A}_{\text{UE},i}(R_i, \text{sde.ct})$.
 - Output $\mathbf{m}'_i \oplus t$.
5. $\mathcal{A}_{\text{SDE}}$ sends $(q+1)$ quantum decryptors $\mathbf{qD}_1, \dots, \mathbf{qD}_{q+1}$ to its challenger.

The challenger of the collusion-resistant identical-challenge search anti-piracy security game samples $s \leftarrow \{0, 1\}^\ell$, generates $\text{sde.ct} \leftarrow \text{SDE.Enc}(\text{sde.pk}, s)$, and runs $\mathbf{qD}_i(\text{sde.ct})$ for all $i \in [q+1]$. In UE, sde.ct works as the decryption key ue.dk of UE. In addition, $\mathcal{A}_{\text{SDE}}$ perfectly simulates ue.ct_i since t is uniformly random. Hence, $\mathcal{A}_{\text{UE},i}(R_i, \text{sde.ct})$ successfully decrypts ue.ct and is expected to output $t \oplus s$ for all $i \in [q+1]$ with non-negligible probability. That is, $\mathbf{qD}_i$ outputs s for all $i \in [q+1]$ with non-negligible probability. This completes the proof. $\square$

Remark 4. By a similar proof, we can also show that the above UE scheme satisfies a variant of multi-challenge search security where each ciphertext encrypts independently random message rather than the same message.

9.3 Multi-Copy Secure Construction

In the UE scheme given in [Section 9.2](#), the encryption algorithm has classically determined outputs ([Definition 6](#)) if the key generation algorithm of the underlying SDE scheme does. As shown in [Section 8](#), there exists a collusion-resistant identical-challenge search anti-piracy secure SDE scheme whose key generation has classically determined outputs assuming the existence of iO and OWFs. Thus, we can apply the compiler of [Theorem 7](#) to the multi-challenge UE scheme. We omit the description of the scheme, as it is very similar to that of the quantum coin scheme in [Section 6.1](#). Thus, we obtain the following theorem.

Theorem 19. *Assuming the existence of polynomially secure indistinguishability obfuscation and one-way functions, there exists an unclonable encryption scheme with multi-copy search security.*

Remark 5. When we apply the compiler of [Theorem 7](#), the keys of the PRS and PRF can be treated as part of the encryption key of the multi-copy search-secure UE scheme, while they are not included in the decryption key so that they are hidden from the adversary. In this way, we can achieve the property that the ciphertext is a pure state fully determined by the encryption key and the message, similarly to the scheme of [\[PRV24\]](#). On the other hand, if we apply the conversion in [Appendix A](#) to make the encryption and decryption keys identical, this deterministic ciphertext property is lost. It remains an open problem to construct a multi-copy search secure UE scheme that simultaneously achieves the deterministic ciphertext property and identical encryption and decryption keys.

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A Making Encryption and Decryption Keys Identical in UE

In the syntax of UE given in [Definition 26](#), the encryption and decryption keys are allowed to differ, which deviates from the standard definition. Here, we show that this discrepancy can be generically resolved while preserving multi-copy security.

Theorem 20. *If a UE scheme with classical key generation algorithm that satisfies multi-copy search security exists, then there also exists a UE scheme that satisfies multi-copy search security and has identical encryption and decryption keys.*

Proof. Let $UE = UE.(KG, Enc, Dec)$ be a multi-copy secure UE scheme. Let ℓ be the length of the decryption key of UE. Then we construct another UE scheme $UE' = UE'.(KG, Enc, Dec)$ as follows.

$UE'.KG(1^\lambda)$:

- Choose $s \leftarrow \{0, 1\}^\ell$.

- Output $ek' = s$ and $dk' = s$.

$UE'.Enc(ek' = s, m)$:

- Generate $(ek, dk) \leftarrow UE.KG(1^\lambda)$.
- Generate $ct \leftarrow UE.Enc(ek, m)$.
- Output $ct' = (ct, dk \oplus s)$.

$UE'.Dec(dk' = s, ct')$:

- Parse $ct' = (ct, t)$.
- Compute $dk \leftarrow s \oplus t$.
- Output $m \leftarrow UE.Dec(dk, ct)$.

It is straightforward to reduce the correctness and multi-copy search security of UE' to those of UE . $\square$

It is easy to see that the multi-copy search secure UE scheme constructed in [Section 9.3](#) has a classical key generation algorithm. Thus, we obtain the following corollary.

Corollary 8. *Assuming the existence of polynomially secure indistinguishability obfuscation and one-way functions, there exists an unclonable encryption scheme with multi-copy search security that has identical encryption and decryption keys.*